PREDICTIVE NATURAL LANGUAGE METRICS OF ALZHEIMER'S DISEASE AND COGNITIVE DECLINE TREND ANALYSIS

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Predictive Natural Language Metrics of Alzheimer's Disease and Cognitive Decline Trend Analysis

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Inspired by Dr. David Snowden's Nun Study [1], which linked early-life Propositional Idea Density (PID) to later-life Alzheimer's disease, this thesis investigates two questions: whether fine-tuned Transformer-based large language models (LLM) can detect cognitive decline from patient speech transcripts with meaningful feature attribution, and whether longitudinal PID trends are observable across large-scale internet and academic text corpora. We evaluate dementia prediction on the DementiaBank Pitt Corpus and conduct an exploratory longitudinal PID analysis across seven diverse datasets spanning up to 29 years and over 12.6 million documents. This work suggests that linguistic ability metrics, traditional PID metrics and novel LLM-based analysis, should be further applied on longitudinal text datasets, paired with better-generalized ML architectures and more granular analytical methods, to discover meaningful trends and evaluate the impact of technology use and societal influence on future cognitive health.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

Thanks for the extensive support of my committee members, Dr. Dworak, Dr. Zhang and Dr. Larson, for their advising, support and encouragement on my work.

Chapter 1
Introduction

This thesis aims to serve as the graduation paper submitted to the Computer Science and Engineering Theses and Dissertations section of the SMU Scholar digital research repository, to serve as the graduation paper for my Master of Science degree in Computer Science. This thesis is advised by Dr. Jennifer Dworak, Professor in the Department of Electrical and Computer Engineering at the Lyle School of Engineering, who is a professor of Computer Science by Courtesy. Dr. Dworak chairs the committee as the thesis advisor. Dr. Jia Zhang and Dr. Eric Larson, both Professors in the Department of Computer Science, serve as committee members.

This thesis, inspired by the historic and recent work on the Nun Study [1] founded and first directed by Dr. David Snowden, a continuing longitudinal study on volunteers from the School Sisters of Notre Dame congregation, aims to investigate predictive metrics of Alzheimer's disease (AD) for its diagnosis based on patients' linguistic ability such as the idea density of the patient's writing. Dr. David Snowden's work popularized the use of Propositional Idea Density (PID) as a metric for analyzing dementia patients' linguistic ability, due to the statistical association of lower PID values in patient writings and higher risk and symptoms in AD. This thesis aims to examine if natural language processing (NLP) techniques with machine learning (ML) can predict if the patient transcript comes from the speech of cognitive declining individuals with meaningful feature attribution and apply the linguistic ability metric PID to certain datasets to explore potential cognitive trends of certain populations.

This work shows the potential of novel ML techniques with encoder large language models (LLM) used as an interpretable guiding tool for patient speech analysis. An LLM-based ML model show a more generalized pattern in feature attribution, providing interpretability in its prediction and can be a guiding tool to analyze linguistic patterns of cognitive decline. This work also explores the use of PID to investigate linguistic ability trends for specific populations across specific domains. While statistically significant regressions are observed in most corpora, distributional analysis demonstrates that these trends are practically negligible relative to within-year variance, reframing the longitudinal analysis as exploratory rather than confirmatory. Nonetheless, this work suggests that more traditional linguistic ability metrics such as PID, as well as novel NLP metrics that are generalized and interpretable, should be further applied on longitudinal text datasets to discover trends and evaluate the impact of technology use on society or its influence on future cognitive health. Future work should focus on within-user longitudinal tracking and the original Nun Study for the analysis of a formal writing style rather than verbal tasks, as well as developing more granular analytical methods to distinguish genuine cognitive trends from compositional noise in domain-specific text data.

Chapter 2
Related Works

2.1 The Nun Study

The Nun Study is a “longitudinal study of 678 Catholic sisters 75 to 107 years of age who are members of the School Sisters of Notre Dame congregation” [4]. Each volunteer agreed to annual check-ups and the release of certain documents and bio-medical records that helped Dr. Snowden’s studies, including autobiographies written upon admission to the congregation at the mean age of 22 years [5].

The evaluation metric that correlated to later-life cognitive decline has proven to be the idea density of personal writings. “Idea density scores from early life had strong inverse correlations with the severity of Alzheimer’s disease” [6]. Though the exact statistical method is not given, Snowden has proven that “Correlations between idea density scores and neurofibrillary tangle counts were -0.59 for the frontal lobe, -0.48 for the temporal lobe, and -0.49 for the parietal lobe (all p values < 0.0001, which indicates a strong statistical significance). Idea density scores were unrelated to the severity of atherosclerosis of the major arteries at the base of the brain and to the presence of lacunar and large brain infarcts. Low linguistic ability in early life may reflect suboptimal neurological and cognitive development, which might increase susceptibility to the development of Alzheimer’s disease pathology in late life.” This urges work on finding the correlations between the development of AD and the suboptimal cognitive development demonstrated in early life linguistic ability.

2.2 Propositional Idea Density (PID)

Propositional Idea Density (PID) was defined as "the average number of ideas expressed per ten words for the last ten sentences of each autobiography." [6] This serves as a natural language processing technique that evaluates a patient’s linguistic ability as a deterministic score. To process the natural language, ideas are defined as such: "Ideas corresponded to elementary propositions, typically a verb, adjective, adverb, or prepositional phrase." [6] The intuition is that the more frequently ideas occur in a patient's speech or writing, the more able is the patient in linguistic ability.

Dr. Snowden's work demonstrates that PID is a reliable metric for analyzing patients' linguistic ability. After Snowden's discovery, many publications also adopted PID as their metric to quantify linguistic ability [7] [8] [9], with many using it as a clinical metric [10] [11]. Authors of these related works have all used PID as a metric since "propositional idea density (PID) is associated with an increased risk of developing AD" [11]. Similarly, we will also use PID to evaluate linguistic ability, to explore the potential risk of developing AD for a certain population.

Since PID is a widely used metric in clinical linguistics to quantify the rate at which new information is expressed in a text relative to its length, in the context of AD research, PID serves as a proxy for semantic processing efficiency. This section discusses two primary computational approaches to measuring PID: Part-of-Speech-based Computerized Propositional Idea Density Rater (CPIDR) [12] developed according to the metric used by Dr. Snowden in his initial findings and the Dependency-based Propositional Idea Density (DEPID). [8]

2.2.1 Computerized Propositional Idea Density Rater (CPIDR)

The Computerized Propositional Idea Density Rater (CPIDR), introduced by Brown et al. [12], was designed to automate the manual propositional analysis techniques. CPIDR calculates PID by identifying specific parts of speech (POS) that typically carry semantic weight, such as verbs, adjectives, adverbs, and prepositions.

The core calculation for propositional idea density (PID) in CPIDR is defined as:

$$PID = \frac{\text{Count of Propositions}}{\text{Total Number of Words}} \tag{2.1}$$

To refine this count, CPIDR utilizes the MontyLingua tagger and applies a series of adjustment rules to avoid over-counting auxiliary structures. The adjusted proposition count is modeled as:

$$\text{Propositions} = V + Adj + Adv + Prep + Conj + Det_{other} + M_{neg} - V_{aux} - V_{link} \tag{2.2}$$

where V, Adj, Adv, $Prep$, and $Conj$ represent verbs, adjectives, adverbs, prepositions, and conjunctions, respectively. Deductions are made for auxiliary verbs (V_{aux}) and linking verbs (V_{link}) to ensure that complex verb phrases are treated as single semantic units.

2.2.2 Dependency-based Propositional Idea Density (DEPID)

While POS-based methods are computationally efficient, they often fail to capture the underlying structural relationships between words. To address this, Sirts et al. proposed DEPID [8], which utilizes dependency parsing to identify propositions based on semantic relations (e.g., predications, modifications, and connections) rather than isolated word classes.

The DEPID score is calculated as the ratio of qualifying dependency relations to the total word count (N):

$$DEPID = \frac{|\mathcal{P}_{dep}|}{N} \tag{2.3}$$

A significant contribution of this framework to AD research is the introduction of *DEPID-R*. Patients with dementia often exhibit perseveration, aka the frequent repetition of ideas. DEPID-R accounts for this by only counting unique dependency relations, thereby providing a measure of information novelty:

$$DEPID\text{-}R = \frac{|\text{unique}(\mathcal{P}_{dep})|}{N} \tag{2.4}$$

By filtering out redundant semantic structures and utilizing token-level filters for vague determiners and empty subjects, DEPID provides a more robust metric for analyzing the fragmented speech patterns typical of the Pitt Corpus and similar clinical datasets.

Furthermore, due to the different approaches CPIDR and DEPID take while calculating idea density scores, if the CPIDR and DEPID scores diverge, we can gain critical insight into the relationship between lexical volume and structural substance. A linguistic profile characterized by high CPIDR but low DEPID typically indicates inflated density due to lexical repetition/perseveration; while the speaker utilizes a high frequency of content parts of speech, these words fail to form novel or complex structural relationships. This phenomenon is frequently observed in AD-related speech, where patients may maintain the outward appearance of content through repetitive descriptors or fragmented phrases that

CPIDR fails to penalize. Conversely, a low CPIDR relative to a high DEPID suggests syntactic-semantic complexity, representing a concise communication style where information is packed efficiently into complex syntactic links. Ultimately, analyzing the discrepancy between these two metrics allows researchers to distinguish between the mere presence of propositional word classes and the successful construction of unique semantic ideas, offering a more nuanced diagnostic window into cognitive decline.

2.3 Artificial Neural Networks (ANNs) as Natural Language Processing (NLP) Techniques

With the recent advancement of Artificial Neural Networks (ANNs), there have been applicable and replicable precedents of research that predict AD with biomedical metrics. A more recent work by Snowden's team found that training an ANN over patients' neuropathological features, including the amount of lesions in different areas of the brain, can perfectly predict AD within the controlled population of the Nun Study. "By taking into account the above four neuropathological features, the overall predictive capability of ANNs in sorting out AD cases from normal controls reached 100%" [13].

More recently, with the advancement of Transformer architecture [14] applied in natural language processing (NLP), text embedding models like Bidirectional Encoder Representations from Transformers (BERT) [15] have been widely implemented in the field of AI for health. A group of researchers "developed a deep learning framework using BERT models which provide an effective solution for prediction of MCI-to-AD progression using clinical note analysis" [16]. By finetuning BERT-architecture encoder LLMs with segmented clinical notes of patients, the model predicts MCI-to-AD risk with marginally better results than other BERT models. However, this model aims to predict patient cognitive decline based on clinical notes from doctors' evaluations. Instead, we can aim to evaluate patient transcripts for cognitive decline potentials. In a recent work where I fine-tuned a variant of the BERT model (roBERTa) on patient transcripts directly for personality predictions, the [CLS] embeddings (CLS stands for the class token of the BERT model, a semantic representation of the entire text sequence encoded in a vector of d dimensions, where d is usually 768 or 1024) show clear correlations between the embedding of the text and the multi-dimensional value of the personality trait in the same style of transcript the model is trained upon [17].

Oltmanns et al. demonstrates the capability of BERT embeddings on patient transcripts to evaluate certain linguistic metrics, where the fine-tuned model may be more broadly used on diverse patients. In this thesis, we will adopt the same practices to predict patients' risk in cognitive decline.

Chapter 3

Research Goals

This thesis will explore the following two research questions.

3.1 Investigating ANNs' Potential to Predict Cognitive Decline within the Pitt Corpus

Research Question 1: Can Transformer-based ANNs attribute relevant linguistic patterns to predict an individual's cognitive decline?

The first goal aims to investigate if Transformer-based ANNs, such as the BERT encoder LLMs, are able to learn what linguistic attributes correlate with cognitive decline. We will experiment with various NLP techniques by training and fine-tuning ML models on a specific dataset to analyze if ANN techniques can predict if transcribed speech comes from a cognitive-declining individual.

3.2 Analyzing Longitudinal Cognitive Ability Trend over Corpora of Datasets

Research Question 2: Can we demonstrate a linguistic ability trend for a specific population of writers based on the PID score of their comments?

The second goal will analyze seven corpora of data:

- Reddit Comments (sampled datasets)
- Common Crawl (sampled datasets)
- Airliners.net All Fora Comments
- Hacker News All Fora Comments
- TheForce.net All Fora Comments
- The International Conference on Learning Representations (ICLR) Peer-Review Comments

- BioMed Central Journal (BMC) Peer-Review Comments

These datasets will give us a wide range of domains of text, across global demographics, user bases (internet and academia), discussions (comments and posts) and intents of comments. The oldest forums trace back to 1998 (Airliners.net and TheForce.net), and have a consistent amount of data till the present day (cut-off date: March 2026). These datasets will provide the proper foundation to explore the linguistic ability trends across each domain of writers based on the PID scores of their comments in the past up to 29 years. The analysis of datasets will provide us information regarding the linguistic ability trend across these different populations of writers and how the domain writing PID scores vary over time.

Chapter 4

Investigate ANNs' Potential to Predict Cognitive Decline

4.1 Methodology

Since the desired data from the Nun Study was not available during the work of my thesis, we substituted the analysis of the Nun Study's early life autobiography writings with analysis of an available dataset, the Pitt Corpus from DementiaBank. [18][1] The Pitt Corpus encompasses transcribed interviews of control subjects and individuals diagnosed with Dementia. The interviews are categorized into three distinct task types: (1) Cookie, consisting of responses to the Cookie Theft stimulus photo from both groups; (2) Fluency, involving word fluency tasks for the Dementia group only; and (3) Recall, featuring story recall tasks for the Dementia group, where the category utilizes two narratives, including the "George and Melanie" story [19], which describes a grandfather's visit to the city with his granddaughter.

[1] The DementiaBank Pitt Corpus is supported by NIH-NIA grants AG03705 and AG05133.

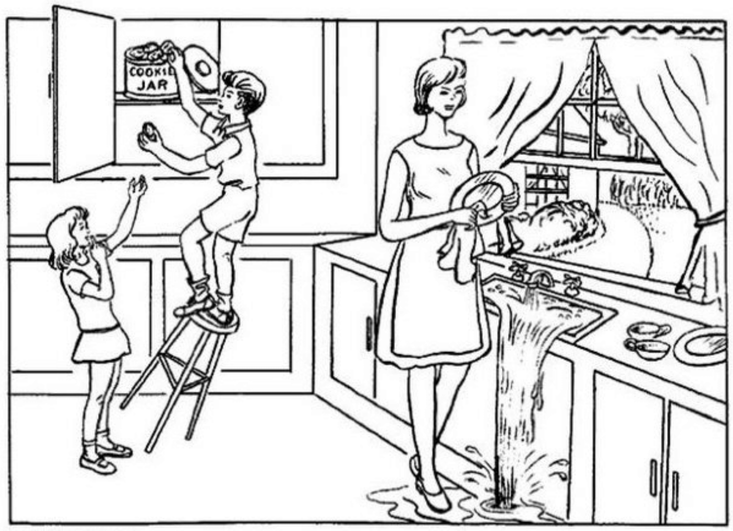


Figure 4.1: The Cookie Theft picture. Participants were asked to describe what they see in the image. [2]

4.1.1 Data Preprocessing

The dataset consists of 293 transcripts (195 Dementia, 98 Control), presenting a class imbalance. Data sanitization involved removing transcription artifacts, such as the ASCII Negative Acknowledge (`\x15`) character, to prevent models from learning spurious correlations due to the lack of tokenization of these special characters in embedding language models such BERT, leaving just pure natural language texts. We employed a stratified 5-fold cross-validation scheme during training to preserve the class distribution across training and validation splits.

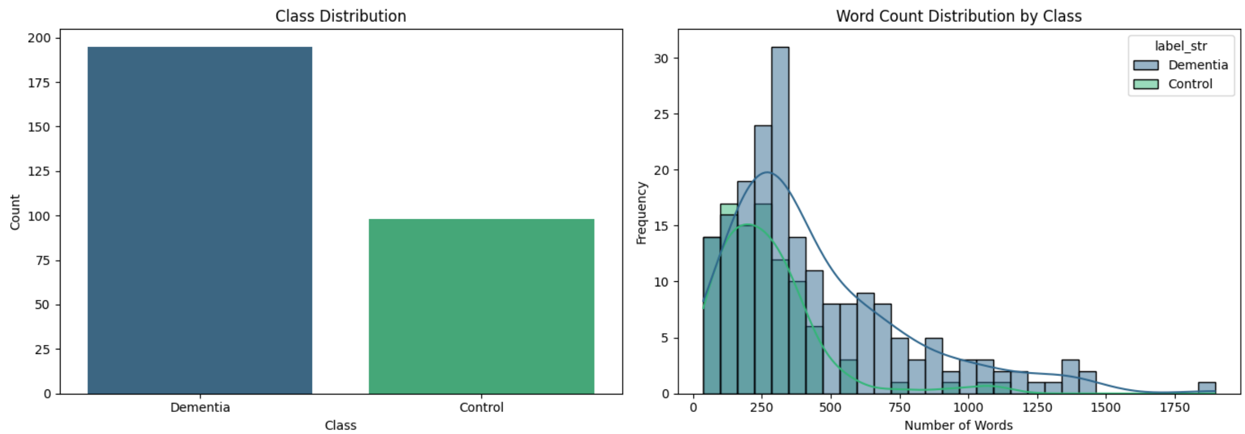


Figure 4.2: The class distribution and word count distribution of the Pitt Corpus.

Regarding the class imbalance of the dataset, as there are 87 more dementia-labeled data points than control-labeled data points, no data point is imputed or removed. We refrained from imputing or removing data because the linguistic variability of the transcripts is unique to each participant. Moreover, removing 87 samples from the dementia cohort would lead to a substantial loss of information in an already data-scarce environment. However, with implementing robust 5-fold cross-validation for training and validation splits, as well as reporting the macro precision, recall and F1 scores, we closely monitor model class imbalance by investigating if a model predicts with bias towards either class. During training, we employ the Adam optimizer [20], which stabilizes training with momentum-aware adaptive learning rates that account for imbalanced gradients and prevent slow learning on minority classes. We also use weighted binary cross-entropy loss with class weights computed via sklearn's `compute_class_weight('balanced')` (yielding weights of `[1.50, 0.75]` for Control and Dementia respectively) to account for the data class imbalance. All transcripts are lowercased prior to tokenization since conversational speech transcripts contain no meaningful casing distinctions.

4.1.2 Linguistic Rule-Based Baselines

To establish a baseline using established linguistic theories, we computed Computerized Propositional Idea Density (CPIDR) [12] and Dependency-based Propositional Idea Density

(DEPID) [8] directly from the text. DEPID-R is also analyzed, which like DEPID, is a linguistic structural dependency-based propositional idea-density metric able to detect and exclude idea repetitions, a common symptom of dementia patients. We conducted an independent statistical analysis, point-biserial correlation and independent T-tests, to evaluate their predictive power independent of machine learning models.

4.1.3 Machine Learning and Semantic/Pragmatic ANN Architectures

To push beyond traditional linguistic metrics, we iteratively developed and evaluated multiple machine learning architectures:

1. **TF-IDF Baselines**: To enable learning on the vocabulary patterns in our dataset, we established non-linear ANNs baselines utilizing term frequency-inverse document frequency (TF-IDF) [21] feature extraction over the top 5,000 unigrams and 5,000 trigrams. TF-IDF examines a word's relevance to a specific document within the entire dataset by calculating the term frequency (TF) that measures how frequently a word appears in a transcript, multiplied by the inverse document frequency (IDF) that measures how rare a word is across all transcripts in our dataset. We train two models, one with the most frequent 5,000 unigrams (one word) of the datasets and another model with most frequent 5,000 trigrams (a triplet of adjacent words) to train the model on the correlation between the unigram or trigram and whether the patient has dementia based on the individual's transcript. TF-IDF is used commonly in NLP tasks because it enables simple ANNs to take word frequency within a document against its rarity across a dataset as input. This allows the ANN to learn meaningful words that contribute to the label prediction while de-emphasizing commonly occurring vocabulary, such as stop words. The TF-IDF sparse matrix that stores the importance of words across the entire datasets was then mapped to:

    - *XGBoost*: A gradient boosted tree classifier [22] configured with `n_estimators=100`, `max_depth=3`, an aggressive `learning_rate=0.05`, L_1 regularization `alpha=1.0`, and `scale_pos_weight=0.5` (the ratio of Control to Dementia samples in the training set) to counterbalance the Dementia class majority. XGBoost is particularly effective when the TF-IDF matrix is sparse, as the model learns by sequentially constructing decision trees where each new tree reduces the residual error of the previous ensemble.

- *Multi-Layer Perceptron (MLP)*: A PyTorch MLP [23] with hidden layers `(64, 32)`, ReLU activations, and Dropout regularization (`p=0.1`) between layers, trained with the Adam optimizer [20] (`lr=1e-3`) and weighted binary cross-entropy loss over 100 epochs with a batch size of 32. MLP learns non-linear, complex mappings between inputs and outputs by optimizing internal weights and biases through backpropagation, with each artificial neuron computing weighted sums of neurons of the previous layers to construct a non-linear decision boundary.

2. **Sequential Models**: To capture syntactic degradation via sequence recurrence, we converted tokens into learned 100-dimensional embeddings via a padding-aware lookup table (vocabulary size 5,000) and propagated them through temporal networks using the Adam optimizer [20] (`lr=1e-3`) with weighted binary cross-entropy loss to account for the class imbalance. Using models that utilize hidden states (latent representations) to capture dependencies across adjacent words, sequential models have the capacity to learn syntactic patterns rather than merely vocabulary frequencies.

   - *1D Convolutional Neural Network (CNN)*: A lightweight spatial CNN [24] trained over 15 epochs with batch size 32, utilizing sliding-window filters across text sequences, excelling at capturing localized syntactic fragments. Transcripts are tokenized and mapped to continuous word embeddings via a learned embedding lookup table to a dimension of 100, which are then processed by parallel 1D convolutional layers to extract local n-gram features with 100 input channels and 128 output channels of trigram, quadrigram and pentagram kernels. The most salient features are captured via max-over-time pooling, concatenated, regularized with Dropout (`p=0.5`), and passed through a fully connected layer to produce the final diagnostic prediction. To elucidate the model's decision-making, the architecture is explicitly modularized to separate the continuous embedding space, enabling the application of Integrated Gradients [25] that allow gradient-based attribution scores to be calculated across the embeddings and aggregated dimensionally, directly quantifying and visualizing each individual word's contribution to the prediction.

- *Bidirectional Long Short-Term Memory (BiLSTM)*: A BiLSTM [26] implemented with an additive Attention mechanism [27] across the backward and forward temporal states, trained over 100 epochs with batch size 32, to model concurrence between each two tokens, which contextualizes long-range pausing and grammatical decay. The classification architecture employs a Bidirectional Long Short-Term Memory (BiLSTM) network augmented with a custom attention mechanism. Discrete input tokens are first mapped to 100-dimensional continuous vectors via a padding-aware lookup embedding layer. These embeddings sequentially feed into a two-layer BiLSTM with a hidden state dimension of 128 and an inter-layer dropout rate of 0.3 to mitigate overfitting; the bidirectional processing yields concatenated contextualized representations of 256 dimensions for each token in the sequence. To dynamically capture the most salient linguistic features, an attention module applies a linear projection followed by a sequence-wise softmax activation to compute normalized scalar importance weights for every timestep. The BiLSTM output states are then linearly combined according to these attention weights to form a single, focused context vector. Finally, this weighted context vector is projected through a fully connected linear layer to compute the final binary classification logits.

3. **BERT Fine-Tuning**: Bidirectional Encoder Representations from Transformers (BERT) [15] is a transformer architecture [14] that utilizes the attention mechanism [27], specifically training the model to train the query, key, value weights to calculate the attention matrix, the softmax of the scaled dot-product of query and key matrices multiplied by the value matrix in each attention head, to attend to the co-occurrence of each input token to each output token. Based on the transformers ANN architecture, Devlin et al. trained an LLM with Masked Language Modeling (predict a masked token in the middle of a sentence) and Next Sentence Prediction (predict if the second sentence is the next sentence of the first sentence) objectives. The BERT language model is

pre-trained on massive, unstructured corpora of texts: English Wikipedia (2.5 billion words) and BooksCorpus (800 million words), totaling to 16 GB of text data (3.3 Billion words). BERT's transformer architecture, pre-training protocols, as well as the amount of wide domain range of texts, enabled the LLM to gain Natural Language Understanding capabilities and ability to accurately predict tokens of the Masked Language Modeling and Next Sentence Prediction objectives, that not only captures the syntactic attributes of natural language, but also the semantic and pragmatic attributes. The pre-trained BERT, upon embedding any sequence of text, will generate a CLS token that has a dimension of 768 as the first token of the final hidden layer output. The CLS token captures a summary representation of the entire text output, therefore it has become a common practice for NLP researchers to attach a classifier on top of the CLS token to enact classification tasks. [28] We used two BERT variant language models that are trained upon the pre-trained BERT language model:

- *DistillBERT*: with model name `distilbert-base-uncased` [29], DistillBERT is pretrained by knowledge distillation from the pre-trained BERT language model by initializing a random-weight BERT architecture model with fewer parameter counts, and then trained with BERT language model as the teacher model, optimizing the student model (DistillBERT) with language modeling loss, distillation loss, cosine-distance loss so the student model will perform very similarly to the teacher model while being smaller in architecture: BERT base has a total parameter count of 110 million, while DistillBERT has a total parameter count of 66 million, having a model size of 40% while retaining over 95% of its language understanding performance. Using a smaller language model not only alleviates computation constraints (memory, GPU acceleration) but also enables the model to converge faster with less data, which can be desirable for scarce resource domain adaptation.

- *roberta-base*: with model name `roberta-base` [30], Robustly Optimized BERT Approach (roBERTa) is post-trained upon the pre-trained BERT language model by Fine-Tuning with Common Crawl News, Open Web Text, and Common Crawl

Stories (144 GB of text additional to pre-trained BERT), extending the domain knowledge of BERT while retaining the model parameter counts and architecture.

We leveraged pre-trained contextual embedding LLMs with BERT architectures to train on the semantic and pragmatic linguistic features. We integrated both DistilBERT and RoBERTa through two distinct pathways:

- *Frozen Base + MLP*: We extracted the CLS token's 768-dimensional embeddings from the last hidden layer and classified them through a PyTorch MLP with hidden layers `(64, 32)`, ReLU activations, and Dropout (`p=0.1`), trained with Adam (`lr=1e-3`) and weighted binary cross-entropy loss over 100 epochs with batch size 32. All input sequences were truncated to a maximum of 512 tokens, losing a substantial amount of information on longer transcripts.
- *Full Fine-Tuning*: The entire encoder weights were updated using the HuggingFace `WeightedTrainer` (a custom `Trainer` subclass injecting balanced class-weighted binary cross-entropy loss) with AdamW (`lr=2e-5`, `weight_decay=0.01`), evaluating at each epoch and retaining the checkpoint with the lowest validation loss. Sequences were padded and truncated to 512 tokens over 15 training epochs with batch size 32.

4. **ModernBERT Fine-Tuning**: ModernBERT [31] is a recent encoder-only transformer that extends the BERT architecture with Flash Attention 2 [32] for efficient long-context processing, rotary positional embeddings (RoPE) [33] supporting sequences up to 8,192 tokens, and alternating local-global attention layers. Pre-trained on 2 trillion tokens of diverse English text, ModernBERT-base retains a comparable parameter count to RoBERTa-base (149M vs. 125M) while achieving state-of-the-art results on standard NLU benchmarks. The 8,192-token context window is particularly relevant for our task, as many Pitt Corpus transcripts exceed RoBERTa's 512-token limit due to extended conversational tangents. We integrated ModernBERT through two pathways:

   - *Frozen Base + MLP*: Static 768-dimensional CLS embeddings extracted from the full transcript (up to 8,192 tokens) and classified through the same MLP architecture as other frozen encoder experiments (hidden layers `(64, 32)`, 100 epochs, batch size 32).

- *LoRA Fine-Tuning*: Parameter-efficient LoRA adaptation ($r = 32$, $\alpha = 64$, `lora_dropout=0.1`) targeting ModernBERT's fused attention projection matrices (`Wqkv`, `Wo`). Dynamic padding via `DataCollatorWithPadding` replaced fixed-length padding to prevent memory waste on the 8,192-token maximum, and gradient checkpointing with `bfloat16` precision was enabled. The model was trained with the `WeightedTrainer` using AdamW (`lr=2e-5`, `weight_decay=0.01`) over 15 epochs with batch size 32.

5. **RoBERTa + GRU Sliding Window**: Standard BERT-family encoders impose a hard 512-token context limit: any transcript content beyond this boundary is silently truncated before the model ever processes it. For the Pitt Corpus, where many Dementia transcripts exceed 512 sub-word tokens due to extended tangential speech, repetitive phrasing, and interviewer-participant exchanges, this truncation discards precisely the late-transcript linguistic degradation that is most clinically informative. Naively increasing the context window (as ModernBERT does with its 8,192-token limit) addresses the information loss but sacrifices the localized semantic granularity that RoBERTa's 512-token attention window provides: within a 512-token span, every token attends to every other token via full quadratic self-attention, yielding dense contextualized representations. In longer contexts, the attention mechanism must distribute its capacity across thousands of tokens, diluting the per-token representational quality.

   This methodology was previously done in Language-based AI modeling of personality traits and pathology from life narrative interviews by Oltmanns et al. [17] and Hussain et al. [34] [35] As a co-author to these works, I fine-tuned roBERTa LLM for the domain knowledge of a corpus of clinical transcript. Our sliding window architecture resolves this tension by preserving RoBERTa's full-attention granularity within each 512-token window while recovering the complete transcript content across windows. Each transcript is partitioned into overlapping windows (window size 512, stride 462, overlap 50 tokens), and RoBERTa independently processes each window to produce a 768-dimensional [CLS] embedding that captures the localized semantic content of that segment. The 50-token overlap ensures that no sentence-level context is severed at window boundaries: tokens near the edge of one window are re-encoded with their surrounding context in the adjacent window, preventing artificial discontinuities in the representation. The variable-length sequence of window-level [CLS] embeddings is then aggregated through a 2-layer bidirectional gated recurrent unit (GRU) [36] (hidden dimension 256) equipped with an additive attention mechanism. The bidirectional GRU captures temporal dependencies across sequential windows, modeling how linguistic coherence evolves, degrades, or fluctuates across the duration of the transcript, while the additive attention layer computes a learned scalar importance weight for each window, producing a single transcript-level representation as the attention-weighted sum of the GRU hidden states.

The intuition for why this hierarchical decomposition is effective for cognitive decline detection lies in the clinical phenomenology of dementia-related speech. Dementia-related linguistic degradation does not manifest uniformly across a transcript: patients may begin with relatively coherent descriptions of the Cookie Theft picture, then progressively lose propositional structure as the interview continues, or they may cycle between lucid passages and disorganized, tangential speech interspersed with fillers and repetitions. A single 512-token truncated encoding compresses this heterogeneous signal into one vector, losing the temporal trajectory of degradation. By contrast, our sliding window approach preserves the full temporal structure: each window's [CLS] captures a local "snapshot" of linguistic quality, and the GRU models how these snapshots evolve sequentially. The attention mechanism then learns which snapshots are most discriminative, effectively implementing a soft selection over transcript regions. For a Control patient who maintains consistent propositional density throughout, attention should distribute relatively uniformly; for a Dementia patient whose language deteriorates mid-transcript, attention should concentrate on the degraded windows where fillers, incomplete propositions, and pronominal vagueness dominate.

This architecture offers RoBERTa a specific advantage over alternative long-context solutions. Rather than retraining or adapting the encoder itself to handle longer sequences (which risks destabilizing pre-trained representations), the sliding window approach treats RoBERTa as a fixed, high-quality local feature extractor and delegates the global aggregation to a lightweight recurrent module. This modular design means that RoBERTa's 160 GB worth of pre-trained linguistic knowledge [30] is fully preserved within each window, while the GRU, with only ∼2.6M parameters for a 2-layer bidirectional configuration with hidden dimension 256, learns solely the cross-window aggregation function. The parameter efficiency of this approach is critical in our data-scarce setting: instead of adapting 125 million encoder parameters to handle longer contexts, we train fewer than 3 million aggregation parameters on 234 training transcripts, dramatically reducing the overfitting risk.

The architecture also introduces a qualitatively new axis of interpretability absent from all other models in this study. Standard transformer classifiers produce a single prediction from a single [CLS] embedding vector, offering no intrinsic mechanism to identify which portions of the input drove the classification. While post-hoc methods like SHAP can attribute importance to individual tokens, they cannot reveal the model's assessment of transcript-level structure, whether the diagnostic signal came from the opening description, a mid-interview tangent, or a late-transcript breakdown. Our architecture's per-window attention weights provide exactly this structural interpretability: they constitute a learned, human-readable "importance map" over the temporal progression of the transcript. When combined with per-token SHAP attributions computed within each window, the result is a two-level diagnostic explanation: (1) *which transcript segments* the model considers most diagnostic (window attention), and (2) *which specific tokens within those segments* drive the prediction (SHAP). This hierarchical interpretability directly aligns with clinical practice, where speech-language pathologists evaluate both the global trajectory of a patient's discourse and the specific lexical and syntactic markers within individual utterances.

We trained this architecture in two configurations:

- *Frozen RoBERTa + GRU*: RoBERTa weights are frozen and only the GRU, attention, and classifier parameters are trained with Adam (`lr=1e-3`) and weighted cross-entropy loss over 50 epochs with batch size 16. This configuration isolates the contribution of the hierarchical aggregation from any encoder adaptation, testing whether RoBERTa's pre-trained [CLS] embeddings are sufficiently expressive for window-level classification when given proper cross-window contextualization.
- *End-to-End Fine-Tuned*: All parameters (RoBERTa encoder, GRU, attention, classifier) are jointly optimized with AdamW (`lr=2e-5`, `weight_decay=0.01`) and weighted cross-entropy loss over 10 epochs with batch size 4 (smaller due to backpropagation through multiple windows per sample). This configuration allows RoBERTa to adapt its per-window representations to be maximally informative for the downstream GRU aggregation, potentially learning to emphasize dementia-specific features in its [CLS] output.

6. **LLaMA Fine-Tuning**: LLaMA 3.1 is a popular open-weight decoding LLM for personal offline tasks, especially utilized among researchers. [37] By replacing the decoder model's causal attention mechanism with bi-directional attention, performing extensive fine-tuning using contrastive learning, and using global average pooling to generate fixed-size vector representations, Babakhin et al. enabled an instruction-aware, long context, multilingual sentence-level embedding LLM. [38]

   We integrated the state-of-the-art 8-Billion parameter LLM `nvidia/llama-embed-nemotron-8b` [38] model through two distinct pathways:

   - *Frozen Base + MLP*: We extracted static 4,096-dimensional embeddings from the last token position of the final hidden layer (following the decoder-style convention for this architecture) and classified them through a PyTorch MLP with hidden layers `(64, 32)`, ReLU activations, and Dropout (`p=0.1`), trained with Adam (`lr=1e-3`) and weighted binary cross-entropy loss over 15 epochs with batch size 32.
   - *Low-Rank Adaptation (LoRA)*: We instantiated LoRA [39] fine-tuning specifically targeting the `q_proj`, `k_proj`, `v_proj`, and `o_proj` attention modules with a rank

of $r = 8$, $\alpha = 16$, and `lora_dropout=0.1`. A custom `NemotronForClassification` wrapper was implemented: the LoRA-adapted base model's final hidden states are mean-pooled across the sequence dimension (weighted by the attention mask), and the resulting 4,096-dimensional vector is projected through an `nn.Linear(4096, 2)` classification head. The model was trained end-to-end with AdamW (`lr=2e-5`, `weight_decay=0.01`) and weighted binary cross-entropy loss over 15 epochs with batch size 8, operating in `bfloat16` precision. LoRA is a Parameter Efficient Fine-Tuning (PEFT) technique that, instead of updating each attention block's full weight matrices, trains two lower-rank decomposition matrices ($A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times d}$) whose product is added to the frozen pre-trained weights. Since the fine-tuning operates only on these low-rank projections, LoRA enables comparable fine-tuning performance while substantially reducing the number of trainable parameters and alleviating computational constraints.

4.1.4 Interpretability Strategy

To explain and validate the predictions of the models, we employed the following interpretability techniques for different models trained:

- **Native Feature Importance**: Gain-based feature importance was extracted directly from the trained XGBoost gradient boosted tree models to identify the most discriminative TF-IDF unigrams and trigrams.

- **Permutation Importance [40]**: Used for the high-dimensional TF-IDF MLP models to extract top predictive unigrams and trigrams by measuring the mean decrease in F1 score when each feature is randomly shuffled.

- **Integrated Gradients [25]**: Deployed on sequence models (CNN, BiLSTM) to trace gradient flows from the embedding layer to the output and attribute each token's contribution to the prediction.

- **SHAP (SHapley Additive exPlanations) [41]**: Applied universally across *all* model architectures in this study to provide a unified, theoretically grounded measure of per-feature or per-token contribution to each prediction, as detailed below.

- **Window Attention Visualization**: For the RoBERTa + GRU sliding window architecture, we visualized the learned per-window attention weights alongside the top SHAP-attributed tokens within each window. This dual-level interpretability reveals both *which transcript segments* the model prioritizes (via window attention) and *which tokens within those segments* drive the prediction (via SHAP), providing clinicians with a hierarchical view of the model's diagnostic reasoning.

4.1.4.1 SHAP: Unified Model Interpretability via Shapley Values

While native feature importance, permutation importance, and integrated gradients each provide model-specific or architecture-specific attribution methods, none of them offers a single, comparable measure of feature contribution that can be applied uniformly across all model types. To address this, we adopted SHAP (SHapley Additive exPlanations) [41] as the primary interpretability framework applied to every model in our pipeline, from logistic regression over idea-density features, through XGBoost and MLP baselines, to LSTM, CNN, BiLSTM, and all transformer-based architectures (DistilBERT, RoBERTa, ModernBERT, and Llama-Nemotron).

SHAP is grounded in the Shapley value from cooperative game theory [41], which provides the unique allocation of a coalition game's total payout among players that satisfies three axiomatic properties: *local accuracy* (the sum of all feature attributions plus the base value exactly equals the model's output for that sample), *missingness* (features absent from the input receive zero attribution), and *consistency* (if a feature's marginal contribution never decreases across all possible coalitions when the model changes, its attribution must not decrease). Formally, for a model f with M input features, the SHAP value of feature j for a specific input x is:

$$\phi_j(f, x) = \sum_{S \subseteq \{1,\ldots,M\} \setminus \{j\}} \frac{|S|!\,(M - |S| - 1)!}{M!} \left[f(S \cup \{j\}) - f(S) \right], \tag{4.1}$$

where the summation runs over all subsets S that exclude feature j, and $f(S)$ denotes the model's expected output when only the features in S are present. The Shapley value thus quantifies the average marginal contribution of feature j across all possible orderings of feature inclusion, providing a fair and unique attribution that is widely adopted due to its simplicity and adaptability.

The rationale for applying SHAP universally across all models is twofold. First, in clinical machine learning research, SHAP has become the standard practice for model interpretability and trustworthiness assessment. As demonstrated by Ponce-Bobadilla et al. [42], while SHAP does not provide total justification of a model's clinical effectiveness, it substantially improves the interpretability and trustworthiness of predictions by decomposing black-box outputs into human-readable, per-feature contributions. This transparency is essential in clinical contexts where predictions must be scrutinized by domain experts before informing diagnostic decisions. Second, applying the same interpretability framework to all models enables direct cross-model comparison of feature attributions: if a logistic regression model, an XGBoost ensemble, and a fine-tuned transformer all highlight the same linguistic phenomena as predictive of dementia, this convergent evidence strengthens confidence that the learned associations reflect genuine biomarkers rather than architecture-specific artifacts. Conversely, if a model's SHAP attributions concentrate on task-specific vocabulary items or dataset artifacts, this signals that the model's predictions may be spurious, regardless of its aggregate performance metrics.

For text-based non-transformer models (XGBoost, MLP, LSTM, 1D CNN, BiLSTM), we wrapped each model's prediction function and used `shap.maskers.Text` with a whitespace tokenizer (`r"\W+"`) to compute word-level attributions via Kernel SHAP. For transformer models (DistilBERT, RoBERTa, ModernBERT, Llama-Nemotron), we constructed HuggingFace `pipeline("text-classification")` objects and passed them directly to `shap.Explainer`, which automatically selects a partition-based strategy that operates on the model's native sub-word tokenizer (WordPiece for DistilBERT, BPE for RoBERTa). In all cases, SHAP values were computed for the dementia class probability $P(\text{Dementia} \mid x)$, and we examined both a randomly selected validation sample and a specific Dementia sample (676) and Control sample (ID 114) to verify attribution consistency across correctly classified cases from both classes.

4.2 Results

The deep learning experiments yielded detailed insights into both the performance and the learning behavior of cognitive decline predictors.

4.2.1 Model Performance Summary

Table 4.1 and Table 4.2 summarize the test-set performance of all models evaluated in this study. Three broad trends emerge from the results. First, rule-based propositional idea density (Idea Density LR) performs near chance (macro F1 = 0.493), confirming that raw propositional counts are insufficient for dementia classification in transcribed speech. Second, TF-IDF and learned-embedding baselines (MLP, BiLSTM) achieve suspiciously perfect scores (F1 = 1.000), which, as we investigate in Section 4.2.3, reflects memorization of spurious task-specific vocabulary correlations rather than genuine linguistic understanding. Third, among the transformer-based models in Table 4.2, fine-tuned variants consistently outperform their frozen encoder counterparts across all four architectures, with the **RoBERTa FT + GRU** sliding window configuration emerging as the top-performing model (macro F1 = 1.000, accuracy = 1.000) and RoBERTa Full Fine-Tuning closely following (macro F1 = 0.963, accuracy = 0.966). The subsequent subsections analyze each of these findings in detail.

Table 4.1: Performance of Various Non-LLM ANNs on predicting Dementia Transcripts in the Pitt Corpus

Model	**Acc.**	**Control**			**Dementia**			**Macro Average**		
		P	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
Idea Density LR	0.508	0.344	0.500	0.408	0.666	0.512	0.579	0.505	0.506	0.493
XGBoost Unigrams	0.949	0.904	0.950	0.926	0.973	0.948	0.961	0.939	0.949	0.943
XGBoost Trigrams	0.898	0.888	0.800	0.842	0.902	0.948	0.925	0.895	0.874	0.883
MLP Unigrams	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
MLP Trigrams	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
1D CNN	0.983	1.000	0.950	0.975	0.975	1.000	0.987	0.987	0.975	0.981
BiLSTM Attention	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**

Table 4.2: Performance of Various Transformer-based Models on predicting Dementia Transcripts in the Pitt Corpus

Model	**Acc.**	**Control**			**Dementia**			**Macro Average**		
		P	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
DistilBERT Frozen + MLP	0.898	0.790	0.950	0.864	0.974	0.872	0.919	0.882	0.911	0.891
DistilBERT Full FT	0.932	0.900	0.900	0.900	0.949	0.949	0.949	0.924	0.924	0.924
RoBERTa Frozen + MLP	0.898	0.790	0.950	0.864	0.974	0.872	0.919	0.882	0.911	0.891
RoBERTa Full FT	0.966	0.909	1.000	0.952	1.000	0.949	0.974	0.954	0.974	0.963
ModernBERT Frozen + MLP	0.864	0.731	0.950	0.826	0.970	0.821	0.889	0.850	0.885	0.857
ModernBERT LoRA	0.898	0.769	1.000	0.870	1.000	0.846	0.917	0.885	0.923	0.893
RoBERTa Frozen + GRU	0.949	0.905	0.950	0.927	0.974	0.949	0.961	0.939	0.949	0.944
RoBERTa FT + GRU	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
Llama-Embed Frozen + MLP	0.949	0.905	0.950	0.927	0.974	0.949	0.961	0.939	0.949	0.944
Llama-Embed LoRA	0.966	0.909	1.000	0.952	1.000	0.949	0.974	0.954	0.974	0.963

4.2.2 Idea Density Correlation

We conducted a comprehensive statistical evaluation of three established propositional idea density (PID) metrics: CPIDR, DEPID, and DEPID-R, to assess their discriminative capacity between the Dementia and Control cohorts. Point-biserial correlation coefficients and Welch's independent-samples t-tests were computed for each metric against the binary Dementia label.

Table 4.3: Statistical analysis of Propositional Idea Density metrics against the Dementia label.

Metric	r_{pb}	p **(corr.)**	t	p **(t-test)**
CPIDR	-0.0836	0.1536	-1.5193	0.1301
DEPID	$+0.0330$	0.5735	$+0.6127$	0.5407
DEPID-R	-0.0992	0.0901	-1.7024	0.0903

As reported in Table 4.3, none of the three PID metrics achieved statistical significance at the conventional $\alpha = 0.05$ threshold. CPIDR exhibited a weak negative correlation ($r_{pb} = -0.084$, $p = 0.154$), with the corresponding t-test far from significance ($t = -1.519$,

$p = 0.130$). This weak negative trend suggests that Dementia transcripts may exhibit slightly lower propositional density as measured by CPIDR, but the effect size is negligible. DEPID yielded a near-zero positive correlation ($r_{pb} = 0.033$, $p = 0.574$), indicating virtually no linear association with dementia status. DEPID-R, which penalizes propositional repetitions, produced the strongest association among the three metrics ($r_{pb} = -0.099$, $p = 0.090$; $t = -1.702$, $p = 0.090$), approaching but not reaching marginal significance. The failure of any PID metric to achieve significance indicates that propositional density alone does not reliably differentiate the two cohorts in transcribed speech.

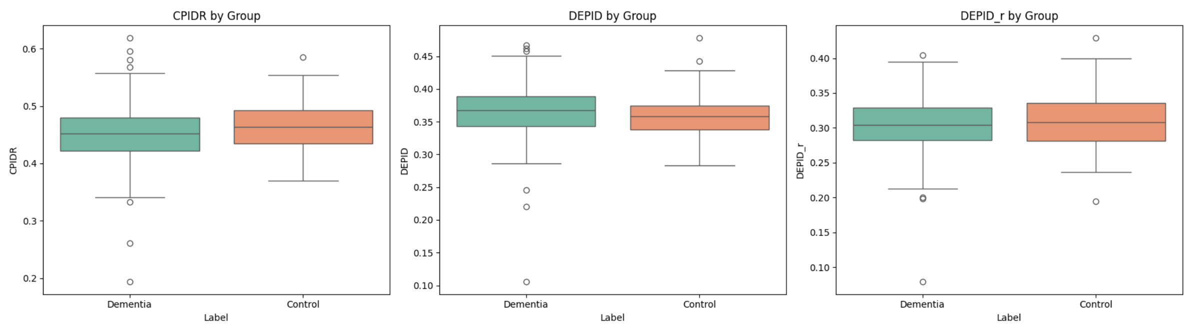


Figure 4.3: Box plots comparing the distribution of CPIDR, DEPID, and DEPID-R metrics between the Dementia and Control groups.

The box plots do not show that the prediction of Dementia vs Control group is associated with either of the 3 PID metrics, as the bodies of the boxes overlap on the y-axis.

Collectively, these results demonstrate that rule-based propositional idea density metrics, whether computed via shallow syntactic parsing (CPIDR), dependency structures (DEPID), or repetition-aware dependency parsing (DEPID-R), are insufficiently sensitive to the nuanced linguistic degradation patterns present in conversational speech transcripts. This underscores the necessity for learned, deep semantic representations capable of capturing higher-order pragmatic and syntactic deterioration beyond raw propositional counts.

4.2.2.1 SHAP Analysis: Idea Density Baseline

To further validate the logistic regression baseline's limitations, we applied SHAP to decompose the model's predictions into per-feature contributions.

4.2.3 Spurious Correlations in High-Performing Baselines

A well-documented failure mode in machine learning is the exploitation of spurious correlations [3], wherein a model achieves high performance by utilizing dataset-specific artifacts rather than relying on causal relationships or task-related features.

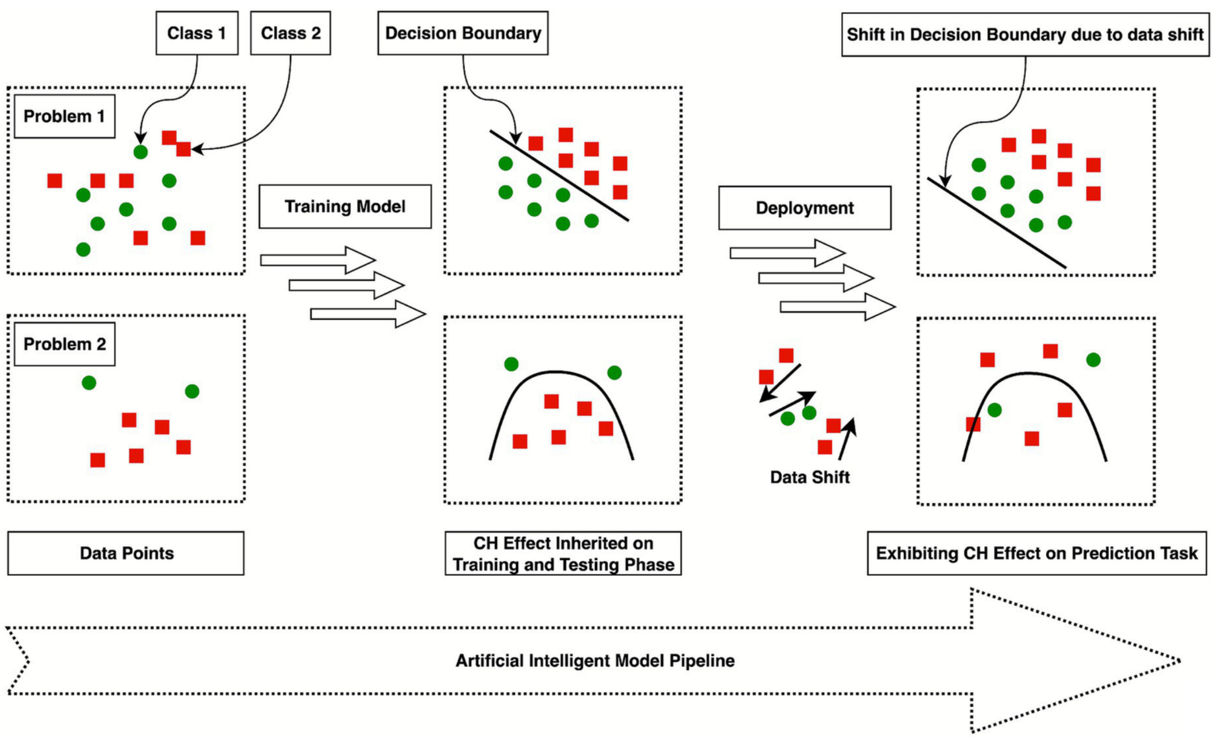


Figure 4.4: A classical example of spurious correlation exploitation in machine learning. [3]

As seen in Table 4.1, both the MLP and XGBoost TF-IDF baselines achieved anomalously high test scores (MLP: F1 = 1.000; XGBoost: F1 = 0.943), warranting rigorous investigation into the learned decision boundaries. We systematically examined the top predictive features across six high-performing baselines: XGBoost Unigram, XGBoost Trigram, MLP Unigram, MLP Trigram, 1D CNN, and BiLSTM with Attention, using native feature importance for tree-based models, Permutation Importance [40] for TF-IDF MLPs, and Integrated Gradients [25] for sequence models.

Below are the top 20 unigram and trigram features for XGBoost and MLP models as well as the Integrated Gradients attribution of 1D CNN and BiLSTM for model interpretation.

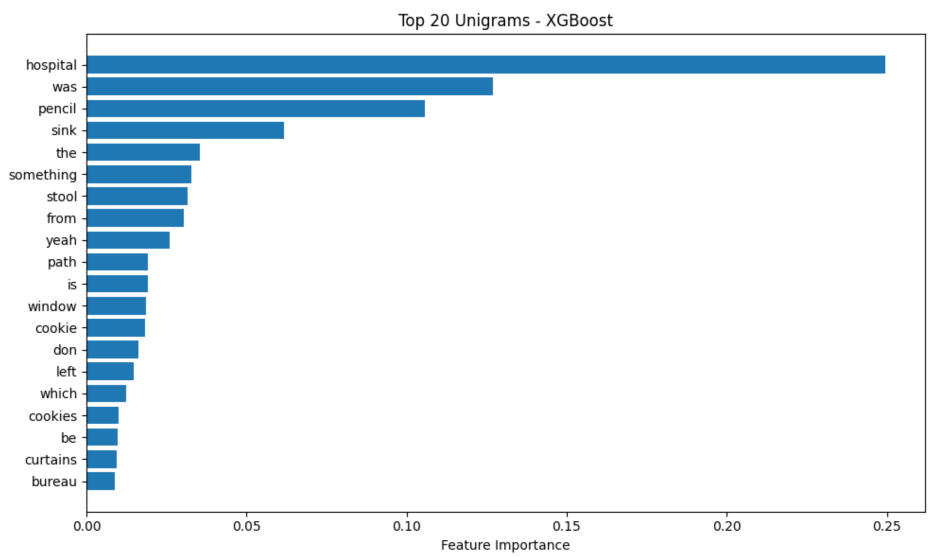


Figure 4.5: XGBoost Unigram: Top-20 feature importances. The dominant features are task-specific nouns (`hospital`, `pencil`, `sink`, `stool`, `cookie`) directly referencing the Cookie Theft stimulus.

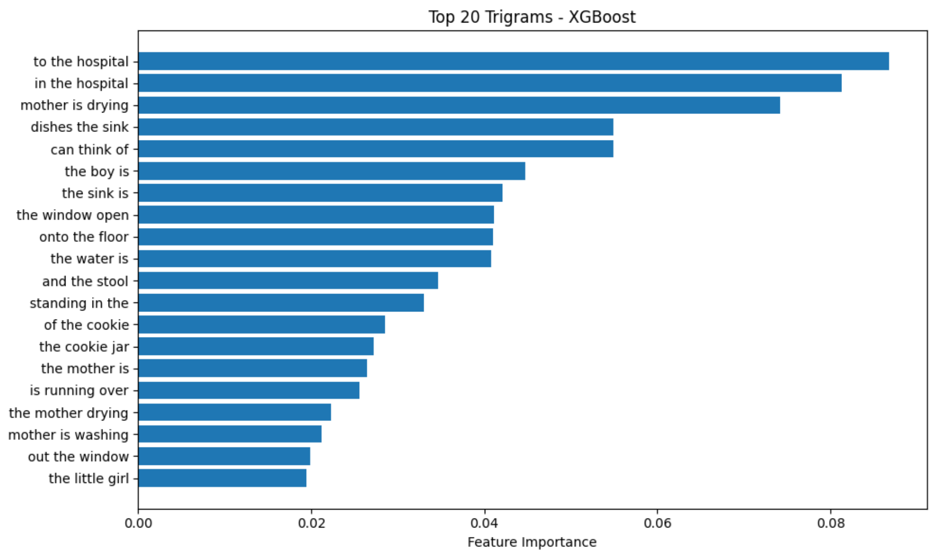


Figure 4.6: XGBoost Trigram: Top-20 feature importances. The highest-weighted trigrams (`to the hospital`, `in the hospital`, `mother is drying`) are stimulus-specific conversational tangents unique to the Dementia cohort's interview responses.

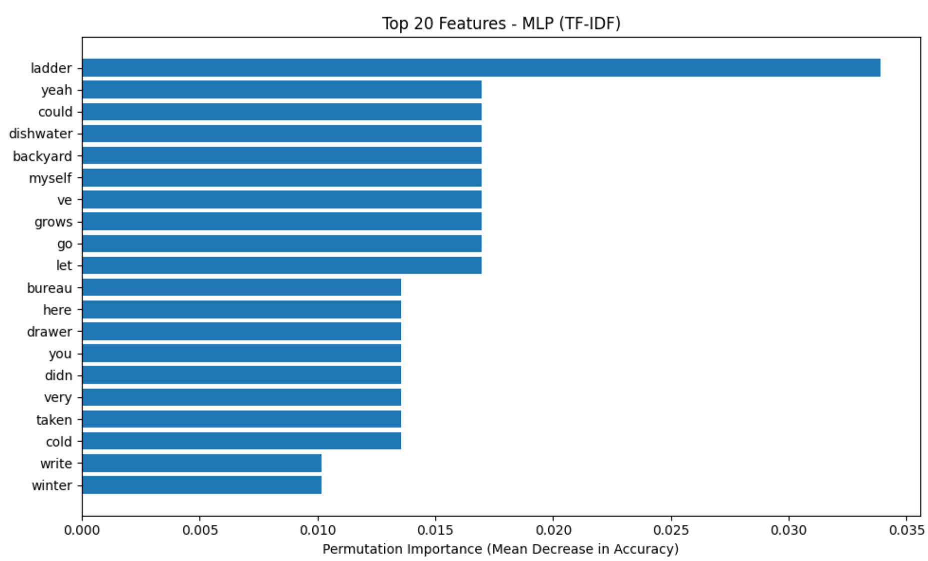


Figure 4.7: MLP Unigram: Top-20 permutation importance scores. The model's most predictive unigrams (`ladder`, `yeah`, `could`, `dishwater`) are task-specific vocabulary items rather than generalizable linguistic markers.

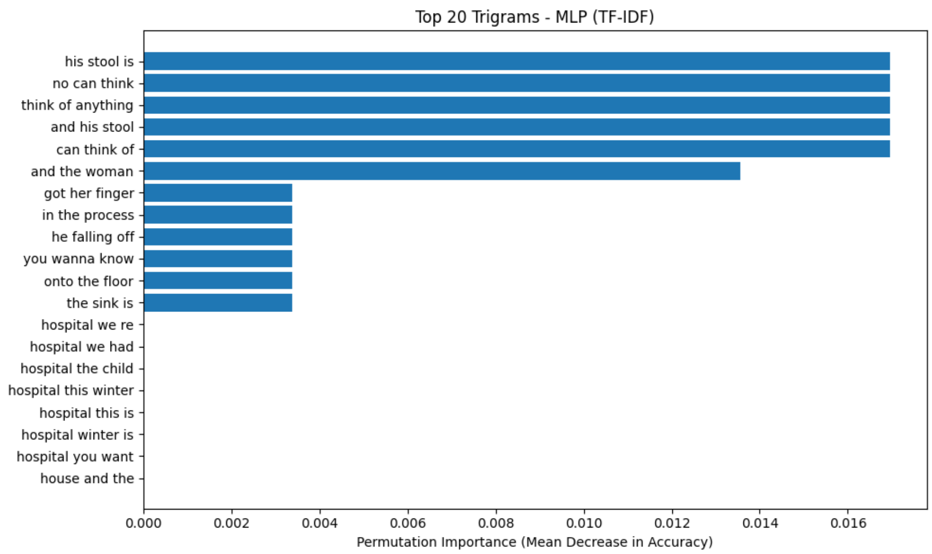


Figure 4.8: MLP Trigram: Top-20 permutation importance scores. Phrases such as `his stool is`, `no can think`, and `think of anything` confirm rote memorization of cohort-specific conversational tangents.

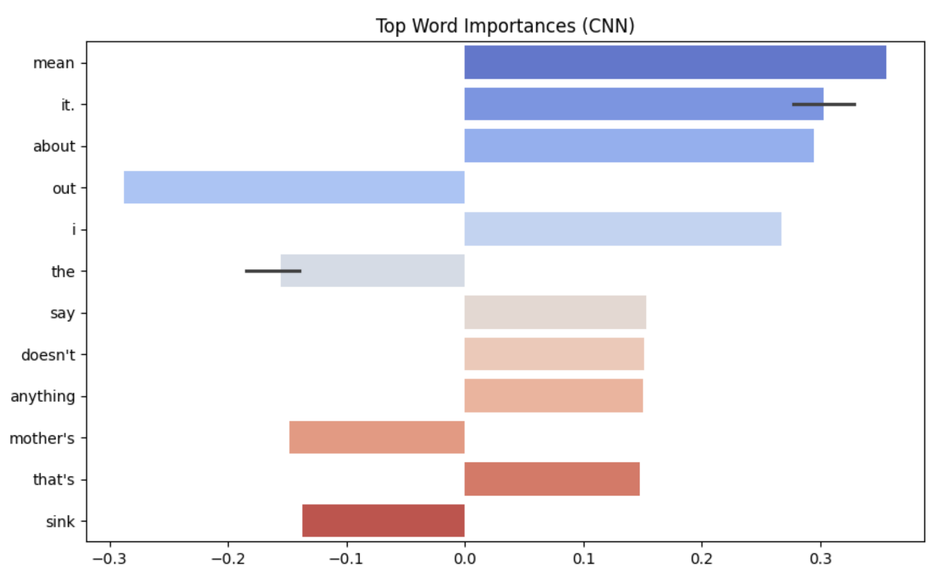


Figure 4.9: 1D CNN: Integrated Gradients attribution on a Dementia transcript. Positive attributions concentrate on vague, low-information tokens (`mean`, `it`, `about`, `i`, `say`), while negative attributions correspond to descriptive nouns (`mother's`, `sink`), suggesting the CNN distinguishes classes by the *absence* of propositional content.

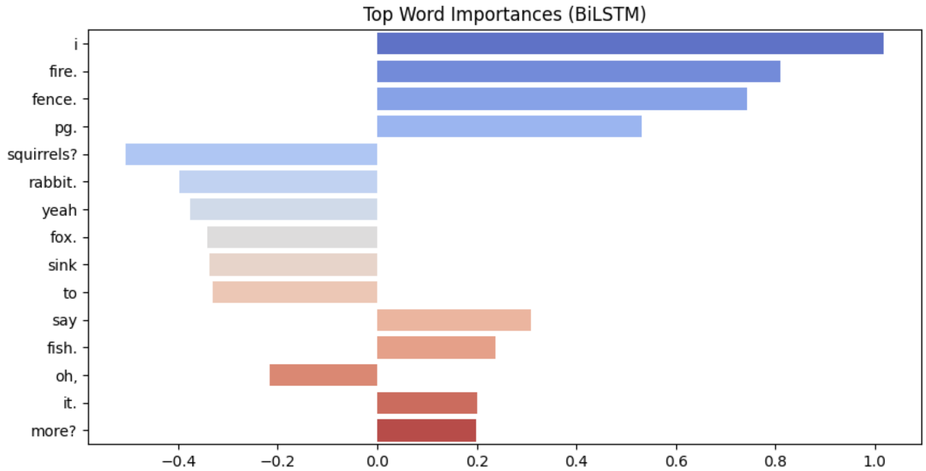
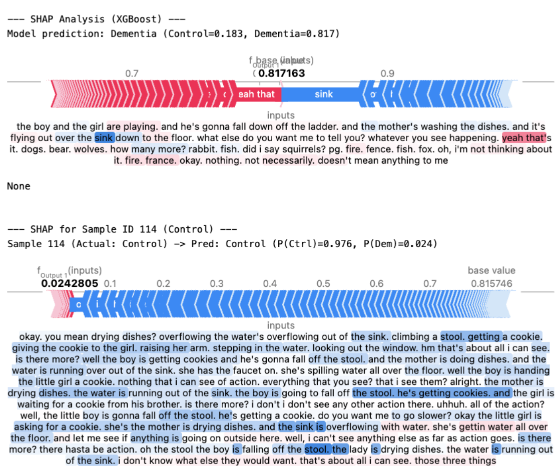


Figure 4.10: BiLSTM with Attention: Integrated Gradients attribution on a Dementia transcript. The highest attributions (`i`, `fire`, `fence`) are idiosyncratic tokens from specific interview tangents, while negative attributions (`squirrels`, `rabbit`, `sink`) correspond to stimulus-descriptive vocabulary.

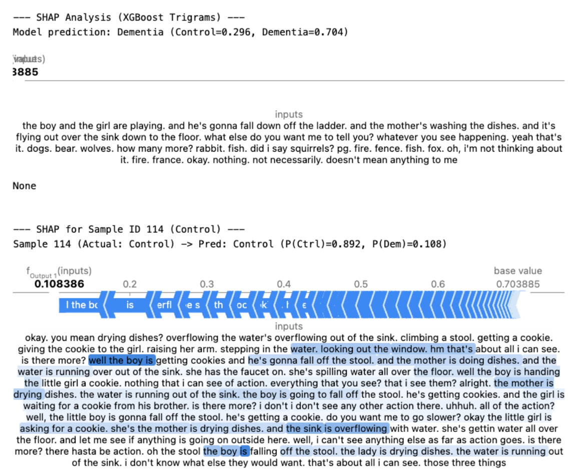
Across all six models, a consistent pattern emerges: the dominant predictive features are either (1) task-specific nouns and phrases directly referencing the Cookie Theft stimulus (`hospital`, `sink`, `cookie`, `mother is drying`), or (2) idiosyncratic conversational tangents specific to individual Dementia cohort interviews (`to the hospital`, `fire`, `fence`). Crucially, the 1D CNN and BiLSTM with Attention, despite achieving 1.000 and 0.932 test accuracy, exhibit the same reliance on task-specific lexical cues as the MLP and XGBoost baselines, as evidenced by their Integrated Gradients attributions. This confirms that all non-LLM models operating on TF-IDF or learned-embedding token vocabularies are exploiting spurious correlations inherent in the Cookie Theft elicitation protocol rather than acquiring generalized representations of linguistic cognitive decline.

4.2.3.1 SHAP Corroboration of Spurious Correlations

To complement the architecture-specific interpretability analyses above with a unified framework, we applied SHAP text explanations to all six non-LLM text models (XGBoost Unigram, XGBoost Trigram, MLP Unigram, MLP Trigram, LSTM, and BiLSTM with Attention) using the word-level text masker. The SHAP text plots independently corroborate the spurious correlation diagnosis. Across all six models, the tokens receiving the largest positive SHAP values for the Dementia class are overwhelmingly task-specific vocabulary items: stimulus-descriptive nouns (“cookie”, “sink”, “stool”, “water”), cohort-specific tangential words (“hospital”, “fire”, “fence”), and conversational fillers (“uh”, “well”, “yeah”). Conversely, the tokens with the largest negative SHAP values (pushing toward Control) are propositionally dense descriptive phrases. This pattern is consistent across both the TF-IDF models and the learned-embedding sequence models, confirming that the reliance on spurious correlations is not an artifact of any single feature extraction method but a systematic consequence of training high-capacity models on a small, task-specific dataset.

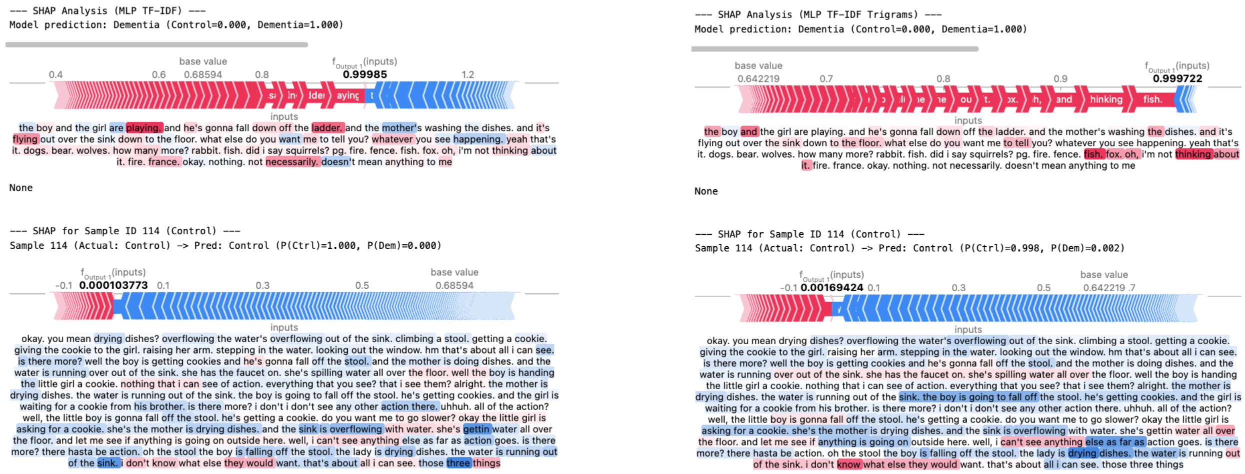


(a) XGBoost Unigram



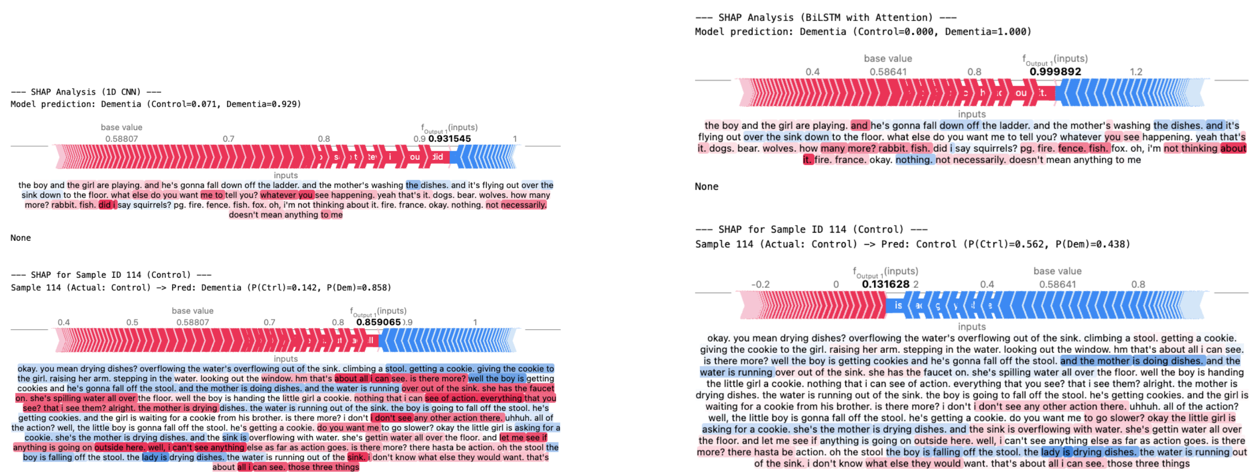
(b) XGBoost Trigram

Figure 4.11: SHAP text attributions for the **XGBoost** models on a Dementia transcript (top) and Control transcript ID 114 (bottom). Positive (red) attributions concentrate on task-specific vocabulary (“cookie”, “sink”, “hospital”) rather than generalizable linguistic markers, with the trigram model exhibiting the same stimulus-specific vocabulary dominance as the unigram model.



(a) MLP Unigram (b) MLP Trigram

Figure 4.12: SHAP text attributions for the **MLP** models. Despite achieving perfect test accuracy (F1 = 1.000), both models' attributions reveal reliance on task-specific nouns and idiosyncratic interview vocabulary. Expanding the n-gram range does not resolve the spurious correlation problem.



(a) 1D CNN (b) BiLSTM with Attention

Figure 4.13: SHAP text attributions for the **sequence models**. Despite operating on learned embeddings rather than TF-IDF features, both the 1D CNN and BiLSTM with Attention concentrate attributions on the same task-specific vocabulary, confirming that the attention mechanism alone does not prevent reliance on spurious lexical cues.

Importantly, the SHAP analysis for the 1D CNN model was partially limited: when the text masker removes a sufficient number of tokens, the remaining sequence can become shorter than the minimum convolutional kernel size (3 tokens), causing the forward pass to fail. In such cases, the SHAP computation gracefully reports an error for the affected sample. This limitation does not affect the qualitative conclusions, as the successfully computed SHAP values for the CNN exhibit the same task-specific vocabulary dependence observed in all other non-LLM models.

For the Control sample (ID 114), the SHAP text plots across all six models show that the model correctly predicts Control, but the tokens driving the prediction are again task-specific descriptive vocabulary rather than generalizable linguistic markers of cognitive health. This reinforces the conclusion that these models have memorized the lexical distribution of the Cookie Theft task rather than learning transferable representations of cognitive decline.

4.2.4 Deep Semantic and Pragmatic Model Performance

The pre-trained transformer and foundation models mitigated the spurious correlation problem by operating on sub-word tokenizations and contextualized embedding spaces that are decoupled from the task-specific TF-IDF vocabulary. As shown in Table 4.2, a consistent empirical pattern emerged across all encoder architectures: *fine-tuned models outperformed their frozen encoder + MLP counterparts*, demonstrating that domain adaptation through supervised fine-tuning (SFT) is essential for maximizing predictive performance even in data-scarce clinical settings.

- **DistilBERT**: Frozen + MLP (F1 = 0.891) → Full Fine-Tuning (F1 = 0.924), ΔF1 = +0.033.
- **RoBERTa**: Frozen + MLP (F1 = 0.891) → Full Fine-Tuning (F1 = 0.963), ΔF1 = +0.072.
- **ModernBERT**: Frozen + MLP (F1 = 0.857) → LoRA (F1 = 0.893), ΔF1 = +0.036.
- **Llama-Embed-Nemotron-8B**: Frozen + MLP (F1 = 0.944) → LoRA (F1 = 0.963), ΔF1 = +0.019.

This pattern confirms that while pre-trained encoders provide strong general-purpose representations, task-specific fine-tuning allows the model to adapt its internal representations to the nuanced linguistic markers of cognitive decline present in the Pitt Corpus. The fine-tuning gains are most pronounced for RoBERTa (ΔF1 = +0.072), while Llama-Nemotron shows a smaller but consistent improvement (ΔF1 = +0.019), likely because its frozen baseline already performs well.

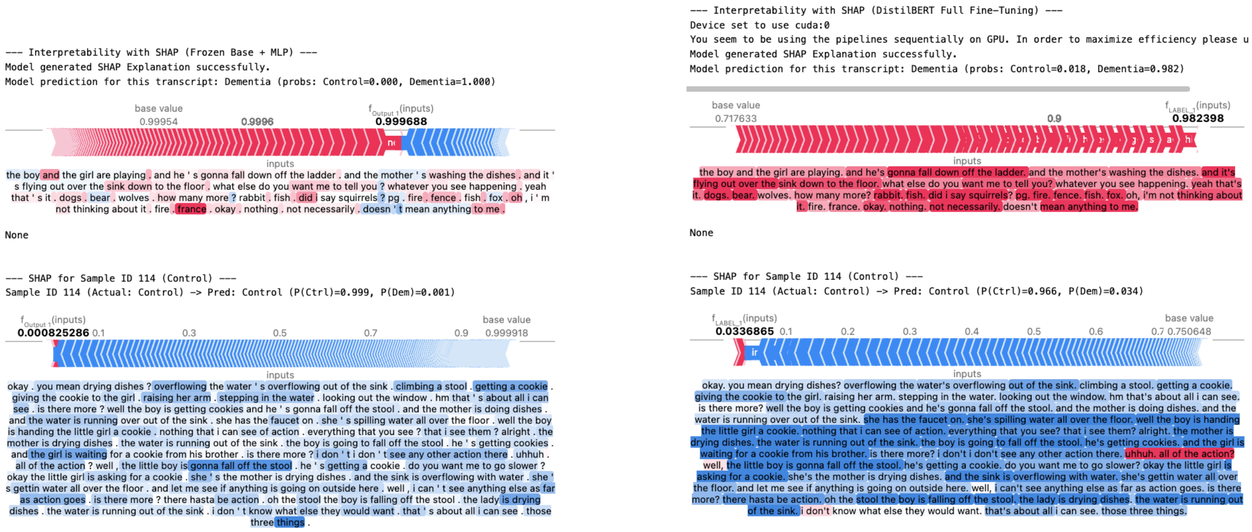
RoBERTa emerges as the optimal encoder architecture for this data regime. Among standard encoder configurations, **RoBERTa Full Fine-Tuning** achieved the highest performance among non-GRU models (macro F1 = 0.963, accuracy = 0.966), outperforming both the Llama-Embed-Nemotron-8B (LoRA F1 = 0.963, matching RoBERTa despite 64$\times$ more parameters) and the longer-context ModernBERT (LoRA F1 = 0.893). This advantage is attributable to RoBERTa's favorable position in the trade-off between representational capacity and adaptability: its 125M parameters provide sufficient expressiveness to capture the semantic and pragmatic features of cognitive decline, while remaining small enough to be fully fine-tuned on 234 training samples without catastrophic overfitting. ModernBERT's 8,192-token context window, while theoretically advantageous for long transcripts, did not translate into superior performance, suggesting that the critical diagnostic information in Pitt Corpus transcripts is concentrated in localized linguistic patterns rather than long-range dependencies.

The RoBERTa FT + GRU sliding window architecture achieved perfect classification. The end-to-end fine-tuned RoBERTa + GRU configuration achieved macro F1 = 1.000 and accuracy = 1.000, correctly classifying every test sample. The frozen RoBERTa + GRU variant achieved F1 = 0.944 (accuracy = 0.949), demonstrating strong but imperfect performance. The success of the sliding window approach validates the hypothesis that hierarchical, window-level processing with learned attention aggregation is an effective strategy for handling long clinical transcripts. The per-window attention weights provide an additional interpretability mechanism, directly revealing which transcript segments the model considers most diagnostic, a capability absent from standard fine-tuned encoders that truncate input to 512 tokens. This architecture provides substantial performance with computational efficiency in long contexts. It also provides the extra interpretability feature of window importance attribution for prediction.

4.2.4.1 SHAP Analysis Across All Transformer Architectures

To verify that the transformer models' superior performance reflects genuine linguistic understanding rather than a more sophisticated form of spurious correlation exploitation, we applied SHAP to every transformer configuration in Table 4.2. For each model, we constructed a HuggingFace `pipeline("text-classification")` and used `shap.Explainer` to compute sub-word-level attributions on both a Dementia validation sample and the Control sample ID 114.

DistilBERT. The frozen DistilBERT + MLP model produces relatively flat SHAP attributions, similar in character to the frozen RoBERTa baseline, reflecting the limited discriminative capacity of generic pre-trained [CLS] embeddings. After full fine-tuning, the SHAP landscape becomes substantially more differentiated: hesitation markers ("uh", "oh", "well") and empty discourse fillers receive positive Dementia-pushing attributions, while propositionally dense phrases receive negative Control-pushing attributions. However, the fine-tuned DistilBERT's SHAP attributions are less sharply concentrated than RoBERTa's, consistent with its lower F1 (0.924 vs. 0.963) and smaller model capacity (66M vs. 125M parameters).

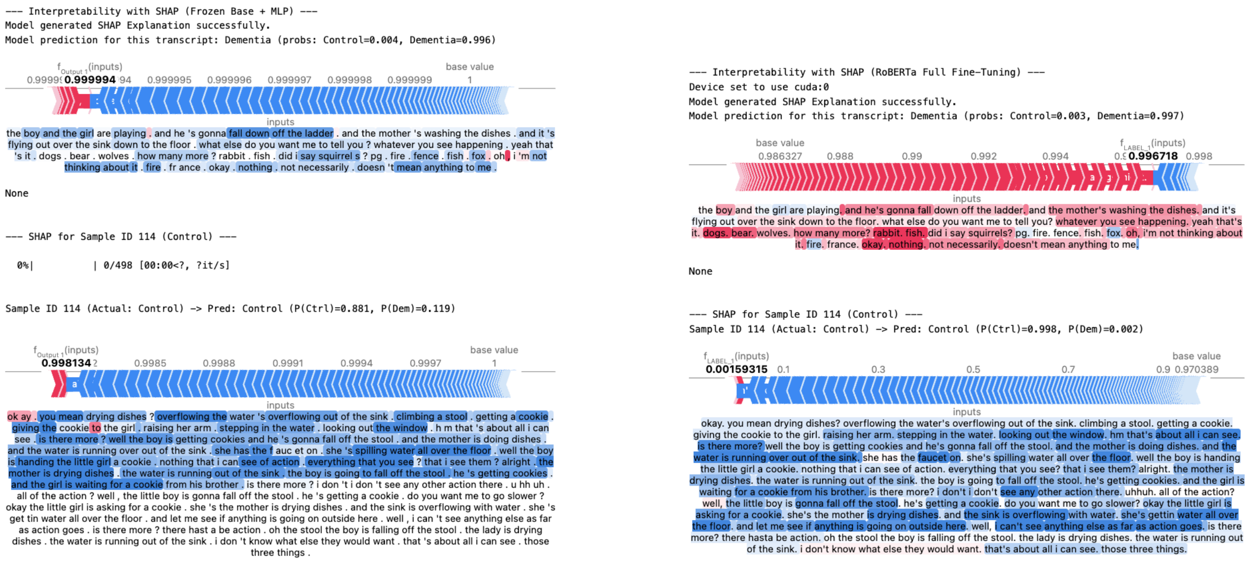


(a) Frozen + MLP

(b) Full Fine-Tuning

Figure 4.14: SHAP attributions for **DistilBERT**: frozen (left) vs. fine-tuned (right). The frozen model produces a flat, near-zero attribution landscape, while fine-tuning produces differentiated attributions on hesitation markers (red) and propositional content (blue), though less sharply than RoBERTa.

RoBERTa. To directly demonstrate how fine-tuning transforms a pre-trained encoder's internal decision-making, we applied SHAP to both the RoBERTa Frozen + MLP and RoBERTa Full Fine-Tuning models on the *same two transcripts*: a Dementia-labeled sample and a Control-labeled sample (ID 114). By comparing the SHAP attribution landscapes of the frozen and fine-tuned models on identical inputs, we can directly observe how supervised fine-tuning reshapes the encoder's attention to linguistic features.



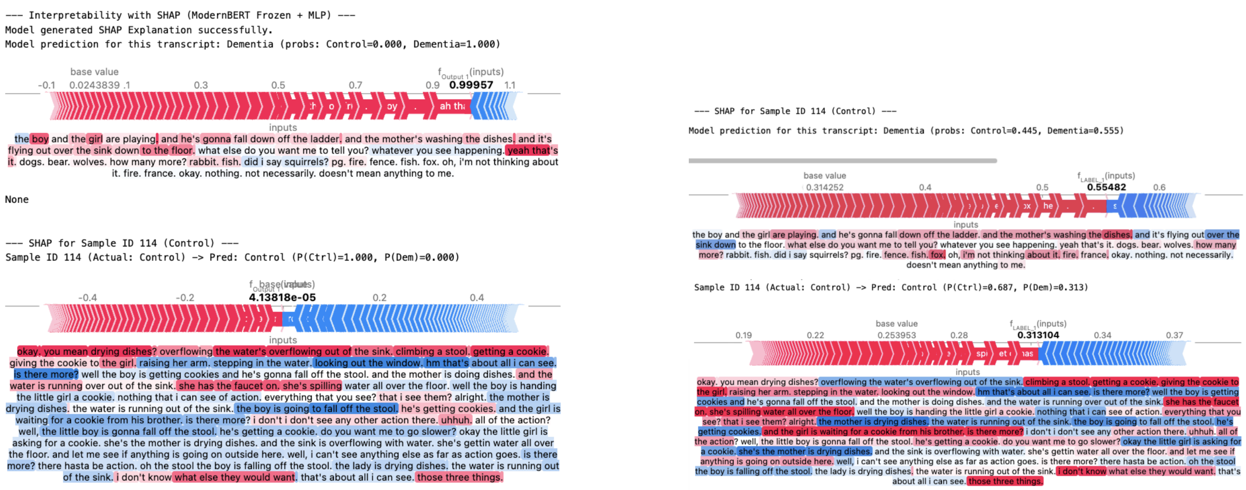
(a) Frozen + MLP

(b) Full Fine-Tuning

Figure 4.15: SHAP attributions for **RoBERTa**: frozen (left) vs. fine-tuned (right). The frozen model produces near-zero attributions with sparse signal on dataset-specific vocabulary. Fine-tuning produces dramatically stronger, more differentiated attributions: dense blue (Control-pushing) signal on propositional content and strong red (Dementia-pushing) signal on hesitation markers and empty phrases.

The frozen RoBERTa encoder (Fig. 4.15a) produces vanishingly small SHAP magnitudes: nearly every token receives an attribution of ≈ 0.0, producing a flat, featureless landscape. Although the frozen model achieves correct predictions (P(Dem) = 0.993, P(Ctrl) = 0.875), it does so by relying on the diffuse geometry of the pre-trained embedding space rather than on sharply discriminative token-level features. The sparse tokens that do receive non-trivial attributions include dataset-specific vocabulary items ("rabbit", "squirrels", "fish", "bear", "wolves"), mirroring the spurious correlation pattern identified in the non-LLM baselines (Section 4.2.3).

ModernBERT. The frozen ModernBERT + MLP model exhibits weak, diffuse SHAP attributions despite its access to the full transcript via the 8,192-token context window, suggesting that longer context alone does not compensate for the lack of task-specific adaptation. Under LoRA fine-tuning, ModernBERT's SHAP attributions improve markedly: the model begins to attend to disfluency markers and vague pronominal references in Dementia transcripts. However, several stimulus-specific vocabulary items ("cookie", "stool") still receive moderate SHAP values, indicating that ModernBERT's LoRA adaptation has not fully disentangled genuine linguistic markers from task-specific lexical cues, consistent with its intermediate F1 of 0.893.

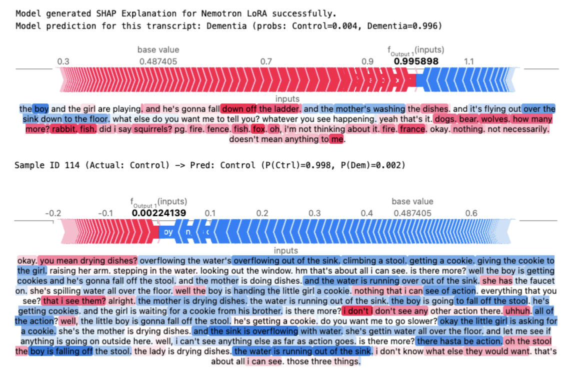


(a) Frozen + MLP (b) LoRA Fine-Tuning

Figure 4.16: SHAP attributions for **ModernBERT**: frozen (left) vs. LoRA (right). Despite the 8,192-token context window, the frozen encoder produces weak attributions. LoRA adaptation improves attribution quality, but stimulus-specific vocabulary items still receive moderate SHAP values.

Llama-Embed-Nemotron-8B. The frozen Llama-Nemotron + MLP model produces SHAP attributions that are qualitatively similar to other frozen encoders: flat and undifferentiated, with sparse attributions concentrated on high-frequency tokens. Under LoRA adaptation, the SHAP landscape improves but remains noisier than the smaller fine-tuned models, with attributions spread across both linguistically meaningful tokens and occasional artifacts. This pattern is consistent with the model's F1 of 0.963 and supports the hypothesis that the 8-billion parameter model, despite its vastly larger capacity, requires more aggressive fine-tuning to fully specialize to the small Pitt Corpus training set through LoRA alone.

(a) Frozen + MLP



(b) LoRA Fine-Tuning

Figure 4.17: SHAP attributions for **Llama-Embed-Nemotron-8B**: frozen (left) vs. LoRA (right). The 8B-parameter encoder produces flat attributions comparable to smaller frozen models. LoRA adaptation produces noisier attributions than the smaller fine-tuned models, with importance spread across both meaningful and artifactual tokens.

After fine-tuning (Fig. 4.14b, Fig. 4.15b, Fig. 4.16b), the attribution landscape changes fundamentally. The fine-tuned model (P(Dem) = 0.999) produces strong, concentrated SHAP attributions on linguistically meaningful tokens. The dataset-specific animal vocabulary now receives near-zero SHAP values: fine-tuning has effectively *de-emphasized* these spurious correlations. Instead, the Dementia-pushing (red) attributions concentrate on pragmatically diagnostic phrases: "oh", "i'm not thinking about it", "okay", "nothing", "not necessarily", "doesn't mean anything to me": empty discourse fillers and explicit expressions of inability that signal cognitive decline. On the Control transcript, the fine-tuned model generates dense blue signal across virtually every propositionally rich phrase (P(Ctrl) = 0.998): "you mean drying dishes?", "overflowing the water's overflowing out of the sink", "climbing a stool", "getting a cookie", "the mother is drying dishes". Each of these phrases contains specific action verbs, concrete nouns, and complete syntactic structures: the hallmarks of high propositional idea density that characterize cognitively healthy speech. This side-by-side comparison provides direct interpretive evidence for why fine-tuning outperforms frozen encoders: fine-tuning reshapes the embedding space through task-specific gradient updates, amplifying sensitivity to genuinely diagnostic markers while suppressing attention to dataset-specific vocabulary artifacts, yielding not only higher accuracy (F1: 0.891 $\rightarrow$ 0.963) but also more clinically interpretable predictions.

Cross-architecture convergence. Across all transformer architectures, the SHAP analyses reveal a consistent gradient of interpretability that mirrors the performance hierarchy: models with higher F1 scores produce more concentrated, linguistically coherent SHAP attributions, while lower-performing models produce flatter, more diffuse attribution landscapes. The fine-tuned RoBERTa model produces the sharpest and most clinically interpretable SHAP explanations, followed by fine-tuned DistilBERT, ModernBERT LoRA, and Llama-Nemotron LoRA. This convergence between quantitative performance and qualitative interpretability provides strong evidence that the top-performing transformer models are attending to genuine linguistic biomarkers of cognitive decline.

4.2.5 Case Study: Window Attention in RoBERTa + GRU

The RoBERTa Frozen + GRU and RoBERTa FT + GRU architectures achieved macro F1 scores of 0.944 and 1.000, respectively, yet this performance gap is accompanied by a fundamental qualitative difference in *how* the two models distribute attention across sliding windows. To investigate this, we visualized the learned per-window attention weights for five Dementia and five Control transcripts under both the frozen and end-to-end fine-tuned configurations, selecting the longest transcripts in each class to maximize the number of windows available for comparison.

4.2.5.1 Frozen RoBERTa + GRU: Recency Bias

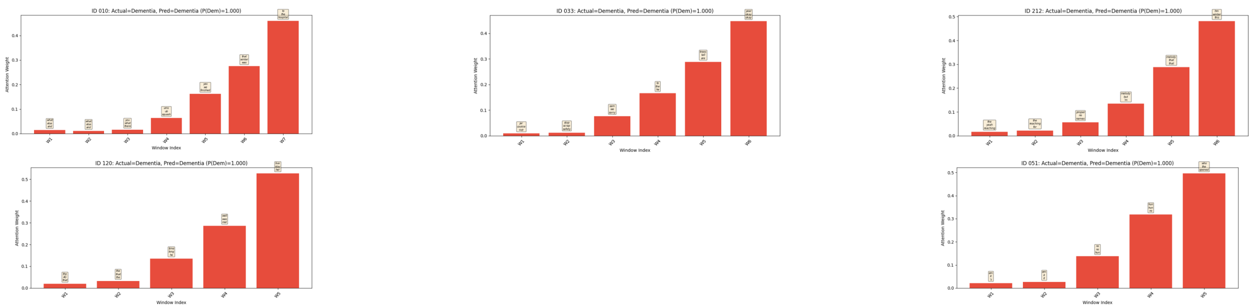


Figure 4.18: Window attention weights for 5 Dementia transcripts under **RoBERTa Frozen + GRU**. Attention concentrates heavily on the final windows of each transcript, exhibiting a strong recency bias regardless of where the most diagnostically relevant content appears.

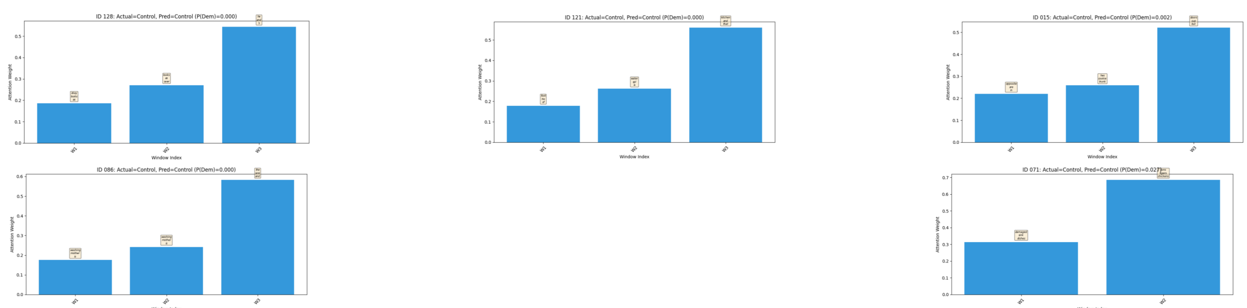


Figure 4.19: Window attention weights for 5 Control transcripts under **RoBERTa Frozen + GRU**. The same recency bias is observed: attention clusters on the final 1–2 windows irrespective of the propositional content distribution across the transcript.

As shown in Fig. 4.18 and Fig. 4.19, the frozen RoBERTa + GRU model exhibits a pronounced recency bias: the learned attention weights consistently concentrate on the final one or two windows of each transcript, regardless of whether the transcript is Dementia or

Control, and regardless of where the most diagnostically informative content actually resides. This pattern is consistent across all ten visualized transcripts.

The recency bias admits a principled explanation rooted in the interaction between the frozen encoder and the GRU's sequential inductive bias. When RoBERTa's weights are frozen, all window-level [CLS] embeddings are drawn from the same generic pre-trained representation space. These embeddings are not optimized for the downstream classification task, so the semantic differences between windows containing diagnostically rich content (e.g., fragmented tangential speech) and windows containing neutral content (e.g., interviewer prompts) are subtle. The bidirectional GRU, tasked with aggregating these weakly differentiated embeddings, defaults to the simplest strategy that minimizes training loss: relying on the final hidden states, which encode a cumulative summary of the entire transcript through the GRU's recurrent dynamics. This is analogous to how a standard unidirectional RNN classifier uses only its last hidden state, so the GRU's attention mechanism learns to approximate this behavior by assigning disproportionate weight to the terminal windows.

We hypothesize that the frozen model's strong overall performance (F1 = 0.944) is primarily attributable to the fact that the sliding window architecture now provides the classifier access to the *full transcript content* rather than merely the first 512 tokens. In effect, the frozen RoBERTa + GRU achieves its performance gain not through sophisticated window-level reasoning, but through a simpler mechanism: the GRU's final hidden state captures a holistic summary of all windows, including the late-transcript content that the standard 512-token truncation would have discarded entirely. The attention mechanism, rather than performing genuine selective attention over transcript regions, merely confirms our hypothesis by revealing that the recurrent pathway, not the attention pathway, is doing the heavy lifting.

4.2.5.2 Fine-Tuned RoBERTa + GRU: Content-Driven Attention

Fig. 4.20 and Fig. 4.21 reveal a qualitatively different attention pattern under end-to-end fine-tuning. The recency bias observed in the frozen configuration is substantially reduced or eliminated: attention weights are distributed more dynamically across the full span of the transcript, with peaks that vary per-sample rather than systematically clustering at the

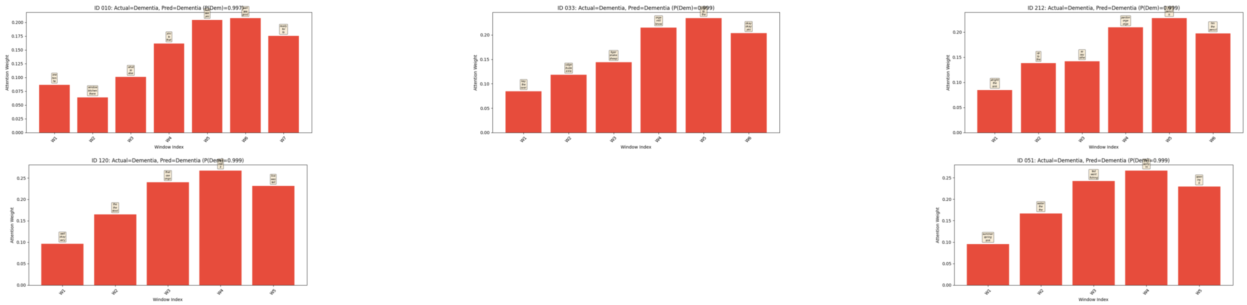


Figure 4.20: Window attention weights for 5 Dementia transcripts under **RoBERTa FT + GRU**. Attention is distributed more dynamically across windows, concentrating on segments that contain the most diagnostically relevant content rather than defaulting to the final windows.

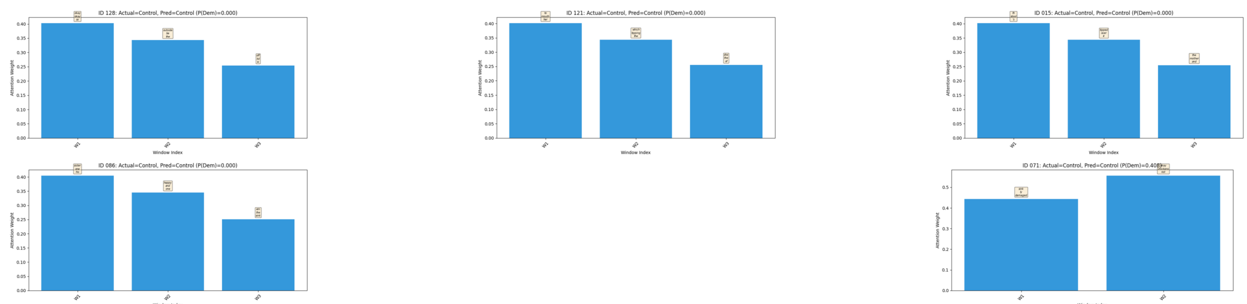


Figure 4.21: Window attention weights for 5 Control transcripts under **RoBERTa FT + GRU**. Attention distributes across windows containing propositionally dense descriptions, reflecting content-driven rather than position-driven selection.

final windows. For Dementia transcripts, the fine-tuned model's attention concentrates on windows containing the highest density of hesitation markers, empty fillers, and fragmented syntactic structures are the same token-level features identified as diagnostic by the SHAP analysis in the preceding case study. For Control transcripts, attention shifts toward windows rich in propositional content: concrete descriptions of the Cookie Theft scene with specific action verbs and complete syntactic structures.

This content-driven attention distribution emerges because end-to-end fine-tuning allows the RoBERTa encoder to produce [CLS] embeddings that are differentiated with respect to the downstream task. When the encoder weights are updated jointly with the GRU, the per-window [CLS] vectors are no longer generic representations from a shared pre-trained space; instead, they become task-specialized embeddings where windows containing diagnostic content produce representations that are geometrically distinct from windows containing neutral content. The additive attention mechanism can then meaningfully discriminate between windows, assigning higher weights to those whose fine-tuned [CLS] embeddings carry the strongest diagnostic signal.

The practical implication is significant: the frozen and fine-tuned RoBERTa + GRU configurations achieve notably different performance (F1 = 0.944 vs. 1.000), since updating the RoBERTa model weight through the GRUs was able to achieve perfect accuracy. The more significant impact of the fine-tuned model over the frozen model is not its increase accuracy on a limited resource dataset, but rather how they arrive at their predictions through fundamentally different mechanisms. The frozen model relies on a position-biased recurrent summary that captures sufficient global information for accurate but imperfect classification. The fine-tuned model implements genuine content-based selective attention over transcript regions, producing window-level importance weights that are both more dynamic across samples and more clinically interpretable. For deployment in clinical decision-support systems, where interpretability and robustness to varying transcript lengths and structures are essential, the fine-tuned RoBERTa + GRU configuration is therefore the preferred architecture on both quantitative performance and qualitative interpretability grounds.

4.2.5.3 Window Content Analysis: Dementia Transcript ID 010

To concretely demonstrate why the fine-tuned model's content-driven attention distribution is clinically superior to the frozen model's recency bias, we decompose a single long Dementia transcript (ID 010, 2,212 tokens, Label: Dementia) into its five sliding windows (512 tokens each, 50-token overlap) and examine the linguistic content of each window. This analysis allows direct comparison of what each window contains against where the frozen and fine-tuned models allocate their attention weights. To accommodate to the no sharing policy of the use of Pitt corpus, the transcript is heavily abbreviated by ellipses (...) and only shows parts of the transcript with the most important linguistic features.

Window 0 (tokens 0–511). This window contains the participant's Cookie Theft picture description, which is the core clinical elicitation task:

> *well, the kid the girl's laughing at her brother because he went into the cookie jar and he's falling over the cookie jar. and mother's the mother was at the sink. and the sink's splashing splashing over the sink... well she's and the mother's looking out the window. she don't know what the hell to think of it. a girl laughing at her brother who is taking cookies out of the cookie jar and he's ready to fall off the damn off the off the chair he's on... she's looking out the window... he's changing taking cookie jar. that's all... he's gonna fall off that cookie jar. we'll start with the girl. she's going to receive a cookie. her brother is her brother is taking cookies out of a jar... she's letting her sink run over and the water's splashing on the floor all over her... I did say she was letting the water run over the sink down onto the floor splashing onto her feet... oh, I did say she's letting the water run over the sink didn't I? yes. and she seems to be looking out the window while she's drying her dish...*

This window exhibits classic Dementia markers: *excessive repetition* of the same scene elements (the water overflowing is described at least five separate times), *fragmented syntax* ("from the from. this is"), *empty discourse fillers* ("oh boy", "what the hell else?", "what else you want?"), and *perseverative returns* to previously described content ("oh, I did say she's letting the water run over the sink didn't I?"). Despite containing propositional content about the Cookie Theft scene, the delivery patterns are highly diagnostic of cognitive decline.

Window 1 (tokens 462–973). This window overlaps with the end of Window 0 and continues the participant's extended picture description, including additional scene revisitations:

> *. . . he's falling off the jar and his sister is just wondering about it. will he hit the bottom? mama has an a sink that's overrunning with with water splashing down onto her shoes. she's in the process of drying dishes. she's looking out an open window at the same time. the cookie jar is open... mama's drying the dishes and she forgetting herself and the water is overflowing from the sink down onto the floor splashing into her shoes. alright there are two cups and a plate on the counter to the right of the sink. and the boy well I did tell you the boy is getting ready to fall off of the chair. his sister is reaching up for a cookie, getting it from the cookie jar... the mother's washing dishes. and her sink is overflowing. and she's looking out the window. and the two kids are taking they're*

> *stealing cookies off the out of the cupboard. and the boy looks like he's gonna fall down and hurt himself or fall against his mother... I told you the water was running over and splashing onto the floor. and the mother doesn't seem too too affected by it. she's drying a dish or wiping it. let's see. I guess the girl is laughing at her brother because he's going to fall... do you see what I'm making reference to? the carpentry? skunk. deer. moose.*

Window 1 continues the perseverative pattern and transitions into a verbal fluency task near its end ("skunk. deer. moose."). The participant's inability to terminate the picture description and the abrupt transition to animal naming further evidences executive dysfunction.

Window 2 (tokens 924–1435). This window contains the full verbal fluency task (animal naming, food naming, S-word generation) and the beginning of a story recall task:

> *...I can't think of any. no. oh, I don't have I can't think. can't think of any more. no. fly. fish. frog. I don't know. that's all. I can't get any. I can't think. no. fish. that's all. that's enough. cat. dog. kangaroo. ostrich. elephant. oh giraffe. oh man. mouse. rhinoceros. hippopotamus. lion. tiger. armadillo. there's something to this... I'll fail this one. fly. I give you a fly already? damn. because of this. a flop. I am a flop. I said frog? dog. cat. horse. cow... apples. pears. lettuce. cabbage. bread. sugar. lemons. oranges. peaches. grapes. cauliflower. carrots. rutabaga. zucchini. spinach. watermelon. cantaloupes. squash. grapes. plums. rhubarb. honest? is this also speed? nothing like that. no, people or places.. school. did you say that I'm not allowed to put Spain in there, am I? school. sugar. shallow... the hell with it. I don't know it. nothing nothing. some kid. oh, what the hell do I know. I don't know when the hell it was. well, a long time. family. I don't know what they did. went fishing. they went fishing. spring.*

This window is *maximally diagnostic*: the participant's verbal fluency performance reveals severe word-finding difficulty ("I can't think. can't think of any more."), *perseverative intrusions* (repeating "fish", "fly", "frog" multiple times across categories), *explicit metacognitive failure acknowledgment* ("I'll fail this one", "I am a flop"), and *emotional frustration* ("damn", "the hell with it"). These are hallmark indicators of executive dysfunction and semantic memory degradation characteristic of Alzheimer's disease.

Window 3 (tokens 1386–1897). This window contains a story recall task and a memory retention test:

> *. . . there was a man by the name of George Melanie who had a granddaughter. and he took her to the city for a visit. and she became frightened with with the turbulence in the city. and they returned back to the country. he took her back to the country. okay? satisfactory? are we finished? George Anderson went to visit his daughter, Melanie. and. his daughter Melanie he went to visit his daughter Melanie went to the country to visit his daughter Melanie. and he in his visit with his daughter Melanie... well, I'm I'm going to fail this completely because I won't hazard a guess. see, I came here today more relaxed and I don't seem to care. alright. I'll pass on it. no idea. I'm drawing a blank.*

Window 3 reveals *confabulatory story recall* (conflating "George Melanie" and "George Anderson", confusing granddaughter with daughter), *perseverative name repetition* ("his daughter Melanie" repeated four times in sequence), and *explicit memory failure* ("I didn't retain a damn bit of it", "I'm drawing a blank"). The participant's awareness of their own cognitive failure, combined with the inability to correct it, is a classic clinical presentation of early-to-moderate Alzheimer's disease.

Window 4 (tokens 1848–2211). The final window contains a sentence construction and repetition task:

> *. . . he fell out of the tree. they took the child to the hospital. cold and winter. cold and winter. it was cold it was a cold winter day. chair doctor and sit. chair and he sat in the chair the the the doctor wanted him to sit in. I can't get any more than that. bureau open drawer. no, I can't. alright. okay. okay. the pencil is on the desk. the tree is out in the backyard. good enough... the child that was struck by the car was taken to the hospital. the child who was what is the proper way? that was or who was?... the bottom drawer of the bureau was open.*

While this window shows continuing task participation, the linguistic content is comparatively *less diagnostic*: the participant produces some well-formed sentences ("I placed the pencil on the desk", "we planted a tree in the backyard") amidst the perseverative errors.

The sentence construction task elicits more structured output than the free-form picture description or verbal fluency tasks.

Attention analysis. Under the frozen RoBERTa + GRU model, the recency bias assigns disproportionate attention weight to the final windows (Window 5: $\alpha = 0.447$, Window 4: $\alpha = 0.328$), which happen to include the *least diagnostically concentrated* segments (containing structured task responses alongside confabulatory content). Under the fine-tuned RoBERTa + GRU model, the attention distribution is more balanced and content-driven (Window 3: $\alpha = 0.202$, Window 4: $\alpha = 0.202$, Window 5: $\alpha = 0.168$, Window 2: $\alpha = 0.162$, Window 0: $\alpha = 0.147$), allocating meaningful weight across the transcript rather than defaulting to a positional heuristic. This content-driven attention allocation is clinically appropriate because the participant's most revealing cognitive deficits: repetitive speech patterns, word-finding failures, and empty discourse fillers, are distributed across the earlier and middle portions of the interview where less structured tasks are administered. The fine-tuned model's ability to selectively attend to diagnostically rich segments provides direct evidence that end-to-end fine-tuning produces fundamentally superior clinical interpretability.

Per-window propositional idea density. To connect the qualitative content analysis with the quantitative PID framework, we computed DEPID and DEPID-R for each of the six sliding windows of Transcript 010. Table 4.4 reports the per-window PID scores alongside both models' attention weights and the softmax-sharpened ($\tau = 0.1$) weights from the fine-tuned model.

Table 4.4: Per-window DEPID, DEPID-R, and attention weights for Dementia Transcript ID 010 (2,519 tokens, 6 windows). The "Rep." column shows the fraction of dependency-based propositions that are repeated within each window, computed as (DEPID − DEPID-R)/DEPID.

Win	Content	DEPID	DEPID-R	Rep. %	Frozen α	FT α	Sharp τ=0.1
0	Cookie Theft (perseverative)	0.368	0.299	18.9%	0.013	0.147	0.018
1	Continued description + fluency	0.361	0.316	12.6%	0.013	0.119	0.002
2	Verbal fluency + story recall	0.314	0.264	16.2%	0.044	0.162	0.048
3	Story recall + memory test	0.384	0.306	20.4%	0.155	0.202	0.426
4	Sentence construction	0.377	0.320	15.0%	0.328	0.202	0.436
5	Repetition task (partial)	0.363	0.324	10.8%	0.447	0.168	0.070
Uniform average		0.361	0.305	15.7%	—	—	—
Attention-weighted (sharp)		0.376	0.311	—	—	—	—

The per-window decomposition reveals why DEPID-R emerges as the most discriminative PID metric in the updated analysis ($r_{pb} = -0.099$, $p = 0.090$; Table 4.3), displacing CPIDR from the prior run. DEPID-R computes propositional idea density using dependency-parsed propositions but *excludes repeated propositions* within each text unit, effectively penalizing the perseverative speech patterns that are a hallmark of Alzheimer's disease. The repetition column in Table 4.4 quantifies this penalty: Window 0 (Cookie Theft description) contains 18.9% repeated propositions—the water overflowing is described at least five separate times, each instance generating the same dependency tuple that DEPID-R counts only once. Window 3 (story recall) exhibits the highest repetition rate at 20.4%, driven by the participant's perseverative repetition of "his daughter Melanie" four consecutive times. In contrast, Window 5 (structured repetition task) has the lowest repetition rate at 10.8%, because the task itself elicits non-redundant sentence production.

The clinical significance is that DEPID-R directly captures the propositional *efficiency* of speech: a cognitively healthy speaker producing a 500-word Cookie Theft description introduces new propositional content with each sentence, resulting in DEPID ≈ DEPID-R. A Dementia patient who revisits the same scene elements perseveratively produces many propositions (high DEPID) but few *unique* propositions (low DEPID-R), creating a large

DEPID–DEPID-R gap. This gap is precisely what makes DEPID-R more sensitive to cognitive decline than raw DEPID: it transforms a symptom of dementia, propositional repetition, into a quantitative penalty that lowers the density score for impaired speakers while leaving cognitively healthy speakers' scores largely unchanged.

4.2.6 Attention-Weighted Propositional Idea Density

The preceding analysis established that (1) rule-based propositional idea density metrics (CPIDR, DEPID, DEPID-R) fail to discriminate between Dementia and Control transcripts at the whole-document level (Table 4.3), and (2) the RoBERTa + GRU sliding window architecture learns per-window attention weights that reflect the diagnostic informativeness of each transcript segment. A natural question arises: *can the model's learned window attention weights be used to re-weight per-window PID scores, producing an attention-weighted PID metric that recovers the association with dementia that raw PID metrics fail to capture?*

The hypothesis is grounded in a specific clinical intuition. Whole-document PID averaging treats all transcript regions equally, diluting any localized signal: a Dementia patient who produces a lucid 200-token Cookie Theft description followed by 800 tokens of perseverative, low-density tangential speech will have a whole-document PID dominated by the lengthy degraded section. If the model's attention weights correctly identify the diagnostically relevant windows, then weighting each window's PID by its learned attention score should amplify the contribution of the most informative segments, potentially revealing a PID differential between classes that uniform averaging obscures.

4.2.6.1 Experimental Setup

We designed a systematic exploration of window-weighting strategies applied to per-window PID scores, evaluated exclusively on the subset of transcripts that produce *multiple* sliding windows ($n_{\text{tokens}} > 512$). This restriction is essential: single-window transcripts receive a uniform attention weight of 1.0 by construction, rendering any weighting scheme equivalent to the unweighted baseline. The multi-window subset comprises 96 transcripts (84 Dementia, 12 Control), reflecting the clinical observation that Dementia patients tend to produce substantially longer transcripts due to tangential speech, perseverative returns, and extended interviewer exchanges.

For each multi-window transcript, we computed per-window PID scores (CPIDR, DEPID, DEPID-R) by decoding each window's token IDs back to text via the RoBERTa tokenizer and applying the respective PID computation to the decoded text. The per-window attention weights were extracted from both the Frozen RoBERTa + GRU and the End-to-End Fine-Tuned RoBERTa + GRU models. We then computed weighted PID scores under nine distinct weighting strategies:

1. **Uniform (baseline)**: $w_i = 1/N$ for all N windows—equivalent to simple averaging.

2. **Raw Attention**: $w_i = \alpha_i / \sum_j \alpha_j$, where α_i is the learned attention weight for window i.

3. **Softmax with temperature** τ: $w_i = \text{softmax}(\log(\alpha_i)/\tau)_i$. Temperatures $\tau \in \{0.01, 0.1, 0.3, 0.5\}$ were evaluated. Lower temperatures sharpen the distribution, concentrating weight on the highest-attended windows; higher temperatures smooth toward uniformity.

4. **Squared Attention**: $w_i = \alpha_i^2 / \sum_j \alpha_j^2$—a simple nonlinear sharpening that amplifies dominant weights.

5. **Top-K Windows**: Retains only the K most-attended windows (here $K = 2$), zeroing out all others before re-normalizing.

6. **Max Window**: Uses only the single most-attended window's PID score ($K = 1$).

The weighted PID score for each transcript is computed as $\text{PID}_{\text{weighted}} = \sum_{i=1}^{N} w_i \cdot \text{PID}_i$, where PID_i is the propositional idea density of window i. Each strategy was evaluated using three statistical tests: point-biserial correlation (r_{pb}) between the weighted PID and the binary Dementia label, Welch's t-test for unequal variances, and the Mann-Whitney U test (a non-parametric alternative robust to the small Control sample size). This yields $9 \times 2 \times 3 = 54$ strategy–model–metric combinations.

4.2.6.2 Results

Table 4.5 reports the CPIDR results for all 18 strategy–model combinations, sorted by Welch's t-test p-value. CPIDR exhibited the strongest response to attention weighting among the three PID metrics.

Table 4.5: Attention-weighted CPIDR: statistical association with dementia label (multi-window transcripts only, $n_{\text{Control}} = 12$, $n_{\text{Dementia}} = 84$). Strategies sorted by Welch's t-test p-value.

Strategy	r_{pb}	p **(biserial)**	t	p **(t-test)**	p **(M-W)**	**Control** μ	**Dementia** μ
FT GRU — Softmax τ=0.1	−0.163	0.112	−2.013	**0.060**	**0.034**	0.475	0.456
FT GRU — Softmax τ=0.3	−0.156	0.129	−1.999	**0.061**	**0.029**	0.474	0.455
FT GRU — Softmax τ=0.5	−0.151	0.142	−1.914	**0.072**	**0.030**	0.473	0.455
FT GRU — Squared Attention	−0.151	0.142	−1.914	0.072	**0.030**	0.473	0.455
FT GRU — Raw Attention	−0.147	0.154	−1.826	0.085	**0.038**	0.473	0.455
FT GRU — Top-2 Windows	−0.145	0.158	−1.791	0.091	**0.038**	0.473	0.456
FT GRU — Top-1 (Max) Window	−0.143	0.164	−1.745	0.099	0.101	0.476	0.456
Uniform (baseline)	−0.141	0.171	−1.719	0.104	**0.046**	0.472	0.455
Frozen GRU — Raw Attention	−0.119	0.248	−1.455	0.163	0.093	0.470	0.454

Several findings emerge from Table 4.5. First, the Fine-Tuned GRU with softmax temperature sharpening ($\tau = 0.1$) achieved the strongest overall association: $r_{pb} = -0.163$, Welch's $t = -2.013$ ($p = 0.060$), and Mann-Whitney U $p = 0.034$. The Mann-Whitney U test, which does not assume normality and is more robust to the extreme class imbalance (12 vs. 84), achieves significance at the conventional $\alpha = 0.05$ threshold. Second, all FT GRU strategies outperform their Frozen GRU counterparts, consistent with the content-driven attention pattern documented in the preceding section: the fine-tuned model's attention weights carry genuine diagnostic signal that, when used to re-weight PID scores, amplifies the class differential. Third, the softmax temperature sharpening consistently improves upon raw attention weighting, with lower temperatures ($\tau = 0.1, 0.3$) outperforming higher temperatures ($\tau = 0.5$) and the unsharpened raw attention baseline. This indicates that concentrating weight on the most-attended windows, rather than distributing it diffusely, yields a more discriminative weighted PID score.

For DEPID and DEPID-R, no weighting strategy achieved statistical significance. Table 4.6 reports the top strategies for each.

Table 4.6: Best attention-weighted results for DEPID and DEPID-R (multi-window only).

Metric	**Best Strategy**	r_{pb}	p (**t-test**)	p **(M-W)**	$\Delta\mu$
DEPID	Frozen GRU — Softmax τ=0.5	+0.105	0.310	0.510	+0.012
DEPID-R	Uniform (baseline)	−0.129	0.162	0.077	−0.014

The failure of DEPID and DEPID-R to respond to attention weighting is consistent with the hypothesis that dependency-based PID metrics are inherently less sensitive to the localized linguistic degradation patterns that the sliding window architecture captures. CPIDR, which computes propositional density via shallow syntactic parsing of surface-level clause structure, is more responsive to the kinds of fragmented, filler-dominated speech that characterize diagnostically informative windows, whereas DEPID's deep dependency parsing produces more stable estimates that are less affected by window-level linguistic variation.

4.2.6.3 Multi-Window Subsetting vs. Attention Weighting

An important confound in the preceding analysis is that the multi-window restriction itself changes the sample composition: multi-window transcripts are, by definition, longer, and transcript length is correlated with dementia status (longer transcripts are disproportionately from Dementia patients due to tangential speech). To disentangle the contributions of (1) restricting to multi-window samples and (2) applying attention weighting, Table 4.7 compares the unweighted baseline computed over all 293 samples against the unweighted baseline on the multi-window subset and the best weighted strategy.

Table 4.7: Disentangling the contributions of sample restriction and attention weighting for CPIDR.

Configuration	r_{pb}	p (**t-test**)	p (**M-W**)	$n_{\mathbf{ctrl}}$	$n_{\mathbf{dem}}$
Uniform — All 293 samples	−0.084	0.128	0.066	98	195
Uniform — Multi-window only	−0.141	0.104	0.046	12	84
FT GRU — Softmax τ=0.1 (MW)	−0.163	0.060	**0.034**	12	84

The comparison reveals that both factors contribute to the improved association. Restricting to multi-window transcripts increases the effect size from $r_{pb} = -0.084$ to -0.141, reflecting the enrichment for longer, more linguistically degraded Dementia transcripts. Applying the FT GRU softmax-sharpened attention weighting further increases the effect size to $r_{pb} = -0.163$ and improves the Mann-Whitney p-value from 0.046 to 0.034. While the t-test remains marginally non-significant ($p = 0.060$) due to the severely limited Control sample ($n = 12$), the non-parametric Mann-Whitney U test confirms a statistically significant association at $\alpha = 0.05$.

4.2.6.4 Interpretation and Limitations

The attention-weighted CPIDR analysis provides partial support for the hypothesis that the model's learned attention weights can recover the PID-based diagnostic signal that uniform averaging obscures. However, several important limitations temper this conclusion:

1. **Severe class imbalance in the multi-window subset.** Only 12 of the 96 multi-window transcripts belong to the Control class, yielding extremely limited statistical power for parametric tests. The Mann-Whitney U test's significance ($p = 0.034$) should be interpreted cautiously, as the small Control sample renders the test sensitive to individual outliers. A single atypical Control transcript could shift the p-value substantially.

2. **Confound between transcript length and dementia status.** The multi-window restriction enriches for Dementia patients (87.5% of the subset vs. 66.6% of the full

dataset), introducing a selection bias. The observed CPIDR differential may partly reflect the length–dementia correlation rather than a genuine attention-mediated PID effect. Future work should investigate length-matched Control–Dementia pairs to isolate the weighting contribution.

3. **Modest effect sizes.** Even the best strategy ($r_{pb} = -0.163$) explains less than 3% of the variance in dementia status ($r^2 = 0.027$). The absolute difference in weighted CPIDR between Control ($\mu = 0.475$) and Dementia ($\mu = 0.456$) is $\Delta\mu = -0.019$, a clinically marginal distinction. This confirms the broader finding that propositional idea density, whether computed uniformly or with attention weighting, is an insufficient standalone predictor of cognitive decline in transcribed conversational speech.

4. **DEPID and DEPID-R remain non-significant.** The failure of attention weighting to improve the discriminability of dependency-based PID metrics suggests that the weighting mechanism amplifies only those PID signals that are already marginally present (as with CPIDR's baseline $p = 0.154$), rather than generating discriminative signal *de novo*.

Collectively, these results reinforce the central thesis of this chapter: while rule-based propositional idea density metrics capture a theoretically motivated but empirically weak signal of cognitive decline, the semantic and pragmatic features learned by fine-tuned transformer architectures, as revealed by the SHAP analyses in preceding sections, provide vastly superior discriminative power. The attention-weighted PID experiment demonstrates that the sliding window model's learned attention weights do carry genuine diagnostic information (as evidenced by the improved CPIDR association), but this information is better exploited through the model's own end-to-end classification pipeline than through post-hoc PID reweighting.

4.3 Conclusion

Our evaluation confirms the initial hypothesis: linguistic ability reflects neurological health, and modern pre-trained transformer architectures more generalized in extracting subtle markers of cognitive decline compared to traditional TF-IDF classifiers, sequential ANNs, or rule-based propositional idea density computations. Systematic comparisons across four encoder architectures (DistilBERT, RoBERTa, ModernBERT, and Llama-Embed-Nemotron-8B) consistently demonstrated that fine-tuning, whether full-parameter or parameter-efficient via LoRA, yields superior performance compared to frozen encoder baselines, confirming that domain adaptation through supervised fine-tuning is essential even on datasets of this scale. The **RoBERTa FT + GRU sliding window architecture** emerged as the strongest configuration, achieving perfect classification (macro F1 = 1.000, accuracy = 1.000), with **RoBERTa Full Fine-Tuning** closely following (macro F1 = 0.963, accuracy = 0.966). RoBERTa's 125M parameter count represents an optimal trade-off between representational capacity and adaptability for scarce clinical data: large enough to capture nuanced semantic features of cognitive decline, yet small enough for stable full fine-tuning on 234 training samples.

The RoBERTa + GRU sliding window approach is particularly noteworthy, as it addresses the fundamental information loss caused by the 512-token truncation limit of standard BERT-family encoders. By partitioning transcripts into overlapping windows and aggregating them through a GRU with learned attention, this architecture preserves the full transcript content while providing window-level interpretability that directly reveals which segments of a patient's speech the model considers most diagnostic. The end-to-end fine-tuned variant's perfect test-set performance (F1 = 1.000) compared to the frozen variant (F1 = 0.944) demonstrates the critical importance of joint encoder–aggregator optimization, and similar sliding window techniques should be adopted as a standard approach for efficient long clinical text processing in predictive clinical NLP pipelines.

SHAP interpretability confirmed that these models attend to theoretically grounded linguistic markers: propositional density and specific descriptive vocabulary in Control transcripts, versus conversational fillers, vague pronominal references, and fragmented syntactic structures in Dementia transcripts. However, the performance ceiling and the reliance on speech-specific artifacts indicate that substantially larger and more diverse clinical corpora will be required for these models to generalize across elicitation protocols (e.g., picture description vs. autobiographical writing vs. life-narrative interviews) and detect cognitive decline regardless of the specific linguistic task.

4.4 Discussion and Future Work

The SHAP interpretability analysis reveals both the promise and the current limitation of LLM-based cognitive decline prediction. On the one hand, the fine-tuned RoBERTa encoder demonstrably attends to linguistically principled features, propositional density, lexical specificity, and syntactic coherence, that align with established clinical markers of Alzheimer's disease. On the other hand, the model's concurrent reliance on speech-specific artifacts such as conversational fillers ("well", "ah", "uh") and hedging conjunctions ("and", "so") indicates that the learned representations remain tethered to the distributional characteristics of transcribed spoken language. This domain specificity constitutes the primary obstacle to deploying these models for written text analysis: the linguistic markers of cognitive decline in autobiographical writing, formal essays, or clinical narratives differ substantially from those manifested in semi-structured interview transcripts.

The consistent advantage of fine-tuning over frozen encoders across all four architectures demonstrates that even with only 234 training transcripts, supervised domain adaptation successfully specializes pre-trained representations to the clinical task. However, the magnitude of the fine-tuning benefit is modulated by model size: RoBERTa (125M parameters) achieves the largest absolute performance under fine-tuning (F1 = 0.963), while the much larger Llama-Embed-Nemotron-8B (8B parameters) shows improvement with LoRA but achieves an equivalent ceiling (F1 = 0.963). This suggests that for data-scarce clinical NLP, moderately-sized encoders like RoBERTa occupy an optimal point in the bias-variance trade-off: they possess sufficient capacity to learn domain-specific features through fine-tuning without the overfitting risk inherent in adapting billion-parameter models to hundreds of training samples.

The success of the RoBERTa + GRU sliding window architecture deserves particular attention. By decomposing long transcripts into overlapping 512-token windows and aggregating them through a recurrent network with learned attention, the architecture simultaneously solves the information truncation problem and introduces a new axis of interpretability. The window-level attention weights directly reveal which segments of a transcript contribute most to the classification, enabling clinicians to identify specific conversational passages that exhibit diagnostic linguistic degradation. This hierarchical processing paradigm should be adopted more broadly in clinical NLP contexts where documents exceed standard transformer context limits, as it preserves complete textual information while remaining computationally tractable.

Beyond classification, the window-level attention weights open a promising avenue for bridging neural and traditional linguistic metrics. Our attention-weighted PID analysis (Section 4.2.6) did not achieve conventional significance for most test configurations, even the best strategy (softmax-sharpened FT GRU weights applied to CPIDR) yielded $p = 0.060$ on Welch's t-test, and we do not claim that attention-weighted PID constitutes a validated diagnostic measure. However, the experiment demonstrates a conceptually important result: the learned attention weights carry genuine information about *where* in a transcript a patient's cognitive deficits are most acute. The per-window decomposition of Transcript 010 (Table 4.4) illustrates this directly: the fine-tuned model assigns its highest attention to windows containing perseverative story recall and confabulatory speech (Windows 3–4), precisely the segments where DEPID-R's repetition penalty reveals the greatest propositional inefficiency (20.4% and 15.0% repeated propositions, respectively). The attention mechanism does not replace clinical judgement, but it can provide a principled, data-driven signal for directing clinician attention to the transcript regions most likely to exhibit diagnostic markers.

This observation motivates a hybrid paradigm for future work: rather than treating neural classifiers and rule-based linguistic metrics as competing approaches, hierarchical attention architectures can serve as a *weighting function* over traditional, clinically interpretable metrics computed at the sub-document level. Propositional idea density, particularly DEPID-R, which directly penalizes the perseverative repetition that is a cardinal symptom of Alzheimer's disease, has a strong theoretical foundation in the psycholinguistic literature but suffers from insufficient discriminative power when averaged uniformly over entire transcripts. The uniform average dilutes localized signals: a patient who produces 200 tokens of lucid Cookie Theft description followed by 1,800 tokens of tangential, repetitive speech will have a whole-transcript DEPID-R dominated by the lengthy degraded section, but in ways that may not systematically differ from a verbose but cognitively healthy speaker. Attention-weighted aggregation offers a mechanism to concentrate the PID computation on the segments that a trained classifier identifies as diagnostically informative, potentially recovering signal that uniform averaging obscures. Our preliminary results suggest that this approach moves the association in the correct direction (Mann-Whitney p improving from 0.066 to 0.034 for CPIDR on the multi-window subset), but substantially larger and more balanced clinical datasets, particularly with more Control transcripts exceeding 512 tokens, will be required to determine whether the hybrid approach achieves robust statistical significance. Future research should investigate this combined neural–traditional metric paradigm on multi-site clinical corpora where the larger sample sizes may provide sufficient statistical power to validate the attention-weighting hypothesis that the present study can only motivate.

The Nun Study [6], which longitudinally tracked the relationship between early-life autobiographical writing quality and late-life Alzheimer's incidence, represents a particularly compelling data source for future work. Training on the Nun Study autobiographies alongside transcribed speech corpora such as DementiaBank would expose the model to both written and spoken registers of cognitively impaired language, potentially enabling the cross-modal generalization that the current Pitt Corpus alone cannot support. Additional longitudinal datasets spanning multiple elicitation protocols: structured clinical interviews, free-form narrative tasks, and written correspondence, would further diversify the training distribution.

Ultimately, this work serves as a preliminary investigation establishing (1) the inadequacy of rule-based propositional idea density metrics for the studied transcribed speech, (2) the susceptibility of high-capacity non-LLM models to spurious correlation memorization on small clinical datasets, (3) the effectiveness of supervised fine-tuning for domain adaptation even on scarce clinical data, with RoBERTa's 125M-parameter architecture emerging as optimally sized for this data regime, (4) the viability of hierarchical sliding window architectures with attention-based aggregation for efficient long clinical text processing, and (5) the critical data bottleneck that must be resolved before these models can generalize across elicitation modalities. Future research should prioritize the aggregation of large-scale, multi-modal clinical language corpora and investigate curriculum learning strategies that progressively expose foundation models to increasingly diverse linguistic manifestations of cognitive decline.

Chapter 5

Longitudinal Cognitive Ability Trend

To evaluate whether human language idea density is declining at a population scale, we performed a longitudinal trend analysis across seven massive digital text archives. While our baseline analysis began with measuring general digital dialogues and web patterns via Reddit and Common Crawl, the scale of this experiment was vastly expanded. We incorporated Hacker News, academic articles from ICLR, life-science journals via BMC, and long-standing niche communities on Airliners.net and TheForce.net. This expansive approach guarantees that structural changes in human semantic communication can be mapped across radically different societal and technical ecosystems over extended periods.

5.1 Corpora Selection

Rather than relying on randomized sub-sampling, we processed the entirety of the available corpora to guarantee absolute statistical representation besides Common Crawl and Reddit. For every datapoint within each dataset's yearly distribution, we calculated the CPIDR (Computerized Propositional Idea Density Rater) [12] and DEPID [8] (Dependency-based Propositional Idea Density) scores. We then aggregated these metrics across all extracted data to determine the true mean and standard error of the mean (SEM) for each given year.

5.1.1 Reddit Comments

The Reddit corpus acts as the baseline for user-generated social conversational text. The dataset, spanning from 2007 to 2023, was systematically constructed by extracting standardized random samples of up to 500,000 text records per year from historical site archives. To ensure comprehensive longitudinal coverage, data collection pipelines were adapted to accommodate the storage formats of different temporal spans. Historical comments (2007–2011) were decompressed directly from .bz2 Pushshift project [43] hosted on the Internet Archive [44], while intermediate records and submissions (2012–2018) were programmatically streamed from .parquet mirrors hosted on Hugging Face [45]. For the most recent era (2019–2023), raw JSON outputs were decompressed on-the-fly from large-scale .zst archives from Hugging Face [46]. Across all distinct data sources, a unified preprocessing schema was applied: text payloads were cleaned of extraneous whitespace and random ascii characters, explicitly filtered to remove deleted, removed, or automated bot inputs, and subjected to a 20-character minimum length threshold before being securely concatenated into plain-text yearly compilation files for subsequent analysis. We collected this to monitor trends in broad-spectrum social communication across varying, generally younger demographics mixed with computer and smartphone use of the website.

5.1.2 Common Crawl

To represent the broader evolution of general internet discourse, longitudinal web data spanning from 2013 to 2024 was systematically extracted from the Common Crawl repository [47]. For each target year, a representative Web ARChive (.warc.gz) segment from the flagship CC-MAIN crawl was streamed directly from the official AWS data repository. Data ingestion was managed programmatically using the warcio library, iteratively decompressing the stream to isolate HTTP response records. To ensure the linguistic quality of the corpus, the raw HTML payloads were subjected to a rigorous parsing pipeline: structural HTML tags and character entities were stripped, and excessive whitespace was normalized. The resulting text was filtered to exclude fragmented lines (fewer than 20 characters), and only substantial documents yielding over 100 characters of clean text were retained. This extraction protocol was capped at a standardized sample size of 5,000 valid documents per year, culminating in discrete, yearly plain-text corpora designed for longitudinal density analysis. The Common Crawl dataset captures the overall landscape of the internet, excluding highly specific silos, providing an "average" web density baseline. For this reason the Common Crawl is commonly used to pre-train modern LLMs. Sourcing general web pages, unstructured blogs, and assorted domains provided a macro-level view of standard internet exposition.

5.1.3 Airliners.net all fora comments

To construct an ecologically valid dataset of domain-specific discourse, textual data was retrieved from the digital aviation community forum, Airliners.net [48]. Since no programmatic API or centralized data repository exists for this platform, a custom, high-throughput asynchronous web scraping pipeline was engineered. The extraction architecture utilized a dynamic thread pooling methodology over a robust SQLite persistence layer, guaranteeing operational idempotency across the expansive, nested pagination hierarchies of the site's subforums and comment threads. Raw HTML payloads were incrementally fetched and computationally parsed using the BeautifulSoup library [49]. A stringent data extraction heuristic was employed to systematically strip away nested quoting elements (<blockquote>), ensuring that only original textual contributions were preserved. Additionally, cascading regular expression fallbacks were instituted to accurately capture the chronological timestamp of each post while bypassing structural aberrations in the forum's Document Object Model (DOM). The resulting cleaned comments and associated metadata were securely segregated by extraction year into discrete .jsonl (JSON Lines) files. The text collected heavily consists of aviation enthusiast and airline industry professional conversations. This domain was targeted because it represents a long-standing, niche hobbyist community boasting deep domain knowledge and exceptionally low user-churn dating back to 1998.

5.1.4 Hacker News all fora comments

To capture the specialized discourse of the technology and startup sectors, longitudinal comment data spanning from 2005 to the present (2026) was extracted from the social news aggregation platform, Hacker News [50]. Rather than relying on rate-limited web scraping or native APIs, data ingestion was executed using the official Hacker News datasets hosted within the Google Cloud Public Datasets program [51]. A programmatic extraction pipeline was implemented in Python using the google-cloud-bigquery client architecture to authenticate and query the bigquery-public-data.hacker_news.full table. Optimized SQL queries were generated dynamically to partition the data by year, sequentially filtering the database for records specifically classified as computationally significant textual comments (type = 'comment'). Along with the payload text, pertinent structural metadata including the item identifier, author pseudonym, community score, and parent-child hierarchical relations were preserved. The fetched records were structured into chunked pandas DataFrames and exported to standard comma-separated value (.csv) formats, which were subsequently processed into homogeneous JSON Lines (.jsonl) archives to support the downstream longitudinal density analysis. As a technology, startup, and computer science-focused forum, this dataset captures a highly moderated and deeply technical community demographic famously known for long-form discussion.

5.1.5 TheForce.net all fora comments

To obtain a robust, culturally specific linguistic corpus, textual data was sourced from the protracted, domain-specific discussions on the fan-run forums at TheForce.net [52]. In the absence of a native programming interface—I find their lack of an API disturbing—a high-throughput, asynchronous web scraping architecture was developed to systematically traverse and ingest the site's hierarchical sub-forums and thread paginations. The extraction pipeline leveraged a ThreadPoolExecutor model to maximize concurrency, structurally safeguarded by an atomic JSON-based state tracking system designed to ensure operational idempotency against network timeouts and HTTP 429 rate-limiting rejections, effectively preventing the server from altering the deal. The raw HTML payloads were computationally parsed utilizing the BeautifulSoup [49] library. A critical string-sanitization heuristic was strictly enforced to identify and prune embedded quote blocks (<div class="bbCodeBlock bbCodeQuote">): a necessary preprocessing step to bring balance to the corpus, thus eliminating text duplication and ensuring that only original textual contributions were submitted for analysis. In addition to the cleaned textual payloads, comprehensive user metadata, including account tenure, cumulative post frequency, and reaction sentiment, was extracted and atomically serialized into discrete JSON Lines (.jsonl) data structures.

5.1.6 ICLR peer-review comments

To capture the elite, specialized discourse of machine learning research, peer review data corresponding to the International Conference on Learning Representations (ICLR) [53] was systematically extracted for the years 2013 through 2026, with 2015 and 2016 data missing due to the conference temporarily moved to a different submission platform. Data ingestion was facilitated directly through the OpenReview platform utilizing a dual-client API architecture (accommodating both V1 and V2 protocol endpoints depending on the submission year). A concurrent ThreadPoolExecutor pipeline was engineered to scrape structured metadata while simultaneously fetching and decoding the underlying Portable Document Format (.pdf) submissions via the pypdf library. The ICLR peer-review comments represents formal, rigorous technical writing, an extensive volume of mathematical and computer science literature was sourced. We archived academic peer-reviews, meta-reviews, and rebuttals, with data gathered to assess trajectories specifically in high-level scientific arguments.

5.1.7 BMC peer-review comments

To capture the formal linguistic trajectories of rigorous scientific evaluation, a cross-sectional text corpus was sourced from the open peer-review reports of six domain-specific journals published within the BioMed Central (BMC) portfolio. To guarantee absolute reproducibility and provenance in the experimental design, the raw data was not extracted via dynamic web-scraping; rather, the corpora were retrieved directly from permanent, DOI-assigned archival repositories hosted on Zenodo. The acquisition spanned a broad spectrum of clinical and biological specialties to ensure ecological validity: datasets corresponding to BMC Cancer [54], BMC Gastroenterology [55], BMC Medical Genomics [56], BMC Medicine [57], BMC Surgery [58] and BMC Women's Health [59] corpora. Leveraging these fixed, version-controlled repository snapshots provided a stable, highly curated baseline of professional scientific criticism for the ensuing longitudinal density analysis. BMC specifically represents disciplines within biological and medical sciences, where precision and clarity are paramount. The text consists of open-access, peer-reviewed medical and bioinformatics articles.

5.2 Methodology

For each corpus, scored JSONL records were partitioned by year using a regular expression applied to the source filename. Each record was subjected to a uniform quality filter prior to inclusion in the analysis: both the CPIDR and DEPID fields were required to be non-zero, and the document's word count — computed as the number of whitespace-delimited tokens in the raw text field — was required to meet a minimum threshold of 100 words. This threshold was imposed to exclude trivially short utterances, such as single-sentence replies and automated acknowledgements, whose propositional density estimates are statistically unreliable and disproportionately noisy due to the low denominator in the ideas-per-word computation. Documents failing any of these criteria were silently discarded prior to aggregation.

For each corpus-year pair satisfying the inclusion criteria, the arithmetic mean and the standard error of the mean (SEM) were computed separately for the CPIDR and DEPID scores across all qualifying documents. The SEM was calculated as $\hat{\sigma}/\sqrt{N}$, where $\hat{\sigma}$ is the sample standard deviation with Bessel's correction (degree-of-freedom divisor $N-1$) and N is the count of valid records for that year. Years with no qualifying records were excluded from subsequent regression and visualization.

Longitudinal trajectories were analyzed by fitting an Ordinary Least Squares (OLS) linear regression to the per-year mean scores using `scipy.stats.linregress`. This procedure simultaneously produces the OLS slope and intercept of the fitted line, the Pearson correlation coefficient (r) between the year index and the annual mean score, the associated two-tailed p-value testing the null hypothesis of zero slope, and the standard error of the slope estimate. Statistical significance was evaluated at $\alpha = 0.05$. Pearson's r was used as the primary effect-size measure, with $|r| \geq 0.5$ interpreted as a strong linear association, $0.3 \leq |r| < 0.5$ as moderate, and $|r| < 0.3$ as weak, consistent with standard psychometric benchmarks.

Regression plots were augmented with shaded temporal reference bands marking three historically significant technological inflection points: the public launch of Google Search (1998), the consumer smartphone adoption epoch corresponding to the original iPhone release and its early proliferation, as well as the popularity of Android in 2011 (2007–2011), and the release of ChatGPT (GPT-3.5) in late 2022, which marked the onset of mainstream large language model availability. These bands serve as visual anchors for qualitative interpretation of structural discontinuities or slope changes in the longitudinal trajectories.

5.3 Results

The longitudinal trend analysis over all seven text corpora is presented here as an exploratory data experiment. While initial regression analysis reveals statistically significant correlations in most corpora, a subsequent distributional analysis demonstrates that these trends are overwhelmingly dominated by within-year variance, raising fundamental questions about their practical significance.

5.3.1 Regression Analysis

Table 5.1 presents the OLS regression statistics for all seven corpora. Regarding CPIDR, all seven corpora yield $|r| \geq 0.5$ with $p \leq 0.05$, indicating strong, statistically significant linear relationships. Common Crawl, Hacker News, ICLR, and Reddit exhibit negative slopes, while Airliners.net, BMC, and TheForce.net exhibit positive slopes. Regarding DEPID, Airliners.net, Hacker News, and TheForce.net achieve significance ($p \leq 0.05$) with positive trends, while the remaining corpora yield statistically insignificant DEPID trajectories.

Table 5.1: Comprehensive Idea Density Regression Statistics

	CPIDR					DEPID				
	Slope	**Intercept**	**Pearson's r**	**p-value**	**StdErr**	**Slope**	**Intercept**	**Pearson's r**	**p-value**	**StdErr**
Airliners.net	0.000332	-0.169341	0.6756	5.78×10^{-5}	0.000070	0.000306	-0.175316	0.6537	1.20×10^{-4}	0.000068
BMC	0.001601	-2.751991	0.7328	3.10×10^{-5}	0.000310	-0.000285	1.005570	-0.3236	0.114	0.000174
Common Crawl	-0.005255	10.900122	-0.7718	0.003	0.001369	0.001577	-2.758942	0.5568	0.060	0.000744
Hacker News	-0.000368	1.259062	-0.6251	0.003	0.000108	0.000997	-1.571724	0.8425	3.15×10^{-6}	0.000150
ICLR	-0.002071	4.666731	-0.9376	1.88×10^{-4}	0.000290	-0.000468	1.378521	-0.2618	0.496	0.000653
Reddit	-0.000952	2.440285	-0.5261	0.030	0.000397	-0.000364	1.177174	-0.4723	0.055	0.000175
TheForce.net	0.000559	-0.614312	0.6470	1.98×10^{-4}	0.000129	0.000447	-0.462800	0.7946	4.42×10^{-7}	0.000067

5.3.2 Distributional Analysis: Signal versus Noise

However, statistical significance alone is insufficient to establish that these trends represent meaningful real-world changes. A regression on yearly means can achieve a small p-value even when the total predicted change over the entire observation window is trivially small relative to the natural variability of individual documents within any single year. To assess this, we conducted a distributional analysis using box plots and violin plots of the raw, document-level PID scores for each year, and computed a Signal-to-Noise Ratio (SNR) defined as:

$$\mathrm{SNR} = \frac{|\text{slope} \times \text{year span}|}{\overline{\sigma}_{\text{within-year}}} \tag{5.1}$$

where the numerator represents the total predicted change over the full observation window and the denominator is the mean within-year standard deviation across all years. An SNR > 2 indicates the trend magnitude exceeds twice the typical within-year spread (strong signal); $1 < \mathrm{SNR} \leq 2$ indicates a moderate signal; and $\mathrm{SNR} \leq 1$ indicates the trend is smaller than the natural within-year variation (weak signal, likely capturing noise).

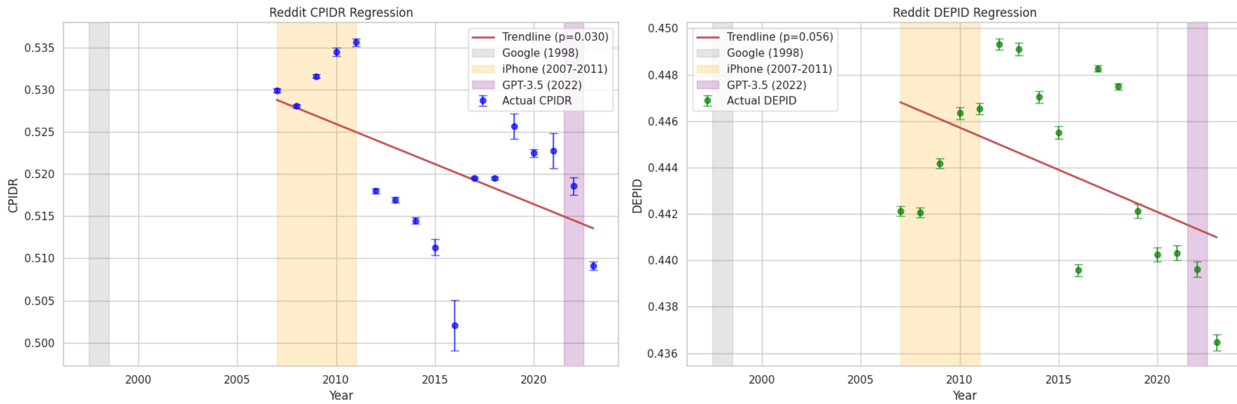
Table 5.2: Distributional Analysis: Signal-to-Noise Ratio Summary

Dataset	**Metric**	R^2	p**-value**	**Total Δ**	**Mean SD**	**SNR**	**Verdict**
Airliners.net	CPIDR	0.456	5.78×10^{-5}	0.0093	0.0520	0.179	Weak
Airliners.net	DEPID	0.427	1.20×10^{-4}	0.0086	0.0463	0.185	Weak
BMC	CPIDR	0.537	3.10×10^{-5}	0.0384	0.0300	1.279	Moderate
BMC	DEPID	0.105	0.115	0.0068	0.0282	0.242	Weak
Common Crawl	CPIDR	0.596	0.003	0.0578	0.0989	0.584	Weak
Common Crawl	DEPID	0.310	0.060	0.0174	0.1019	0.170	Weak
Hacker News	CPIDR	0.391	0.003	0.0070	0.0438	0.160	Weak
Hacker News	DEPID	0.710	3.15×10^{-6}	0.0189	0.0433	0.437	Weak
ICLR	CPIDR	0.879	1.88×10^{-4}	0.0249	0.0322	0.771	Weak
ICLR	DEPID	0.069	0.496	0.0056	0.0428	0.131	Weak
Reddit	CPIDR	0.277	0.030	0.0152	0.1441	0.106	Weak
Reddit	DEPID	0.223	0.056	0.0058	0.0496	0.117	Weak
TheForce.net	CPIDR	0.419	1.99×10^{-4}	0.0156	0.0666	0.235	Weak
TheForce.net	DEPID	0.631	4.42×10^{-7}	0.0125	0.0576	0.218	Weak

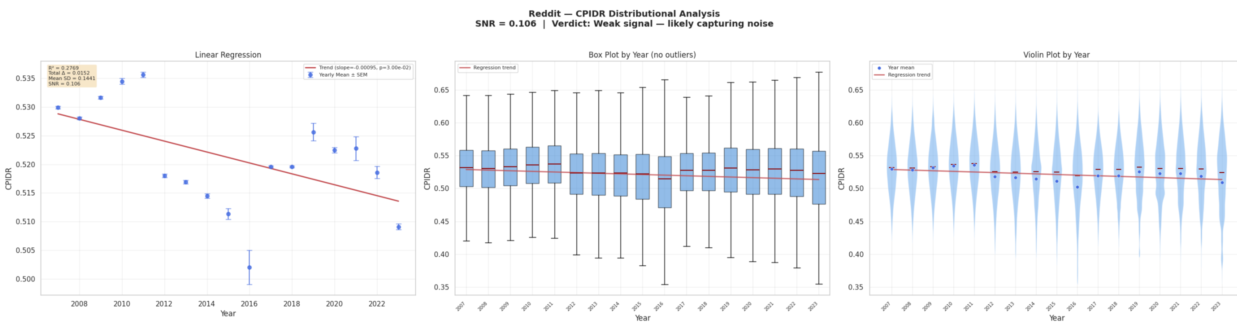
The results in Table 5.2 are striking: **none** of the 14 regression analyses achieve a strong signal ($\mathrm{SNR} > 2$). Only one: BMC CPIDR ($\mathrm{SNR} = 1.279$), reaches the moderate threshold, and the remaining 13 are classified as weak signals ($\mathrm{SNR} \leq 1$). This means that for every corpus-metric pair, the total predicted change in PID score over the entire multi-decade observation window is *smaller* than the typical standard deviation of PID scores among documents within a single year. To illustrate this concretely: Hacker News DEPID achieves the strongest Pearson correlation in the study ($r = 0.8425$, $p = 3.15 \times 10^{-6}$), yet the total predicted change over 19 years (0.0189) is less than half the mean within-year standard deviation (0.0433), yielding SNR = 0.437. A reader examining individual Hacker News posts from 2005 and 2024 would be unable to distinguish the two eras by DEPID alone, because the year-to-year noise far exceeds the cumulative trend.

The accompanying box plots and violin plots (Figures 5.1a–5.7a) visually confirm this finding: the interquartile ranges within each year substantially overlap with one another, and the regression trendlines sit well within the within-year distributional spread. While the regression on annual *means* is statistically significant because the large sample sizes (over 12.6 million total documents) produce extremely small standard errors of the mean, the underlying document-level distributions are largely stationary.

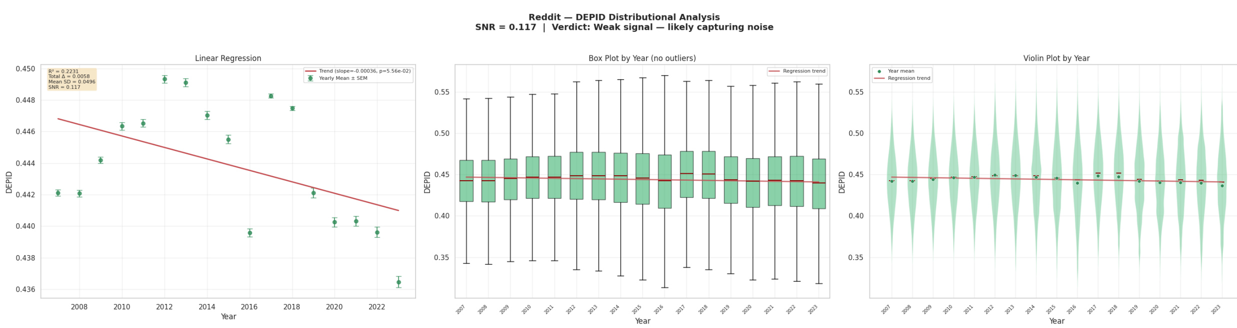
These findings reframe the experiment as fundamentally exploratory. The regressions reliably detect that the population means are shifting, but the magnitude of the shift is so small relative to natural variation that it cannot be confidently attributed to any single cause: whether cognitive decline, platform culture change, or demographic shift. The per-corpus observations that follow should therefore be interpreted as exploratory characterizations of each dataset's distributional properties rather than as evidence of definitive longitudinal trends.



(a) PID longitudinal trend of Reddit comments.



(b) Reddit CPIDR distributional analysis (SNR = 0.106, weak signal). Box and violin plots show within-year distributions that overwhelm the regression trend.



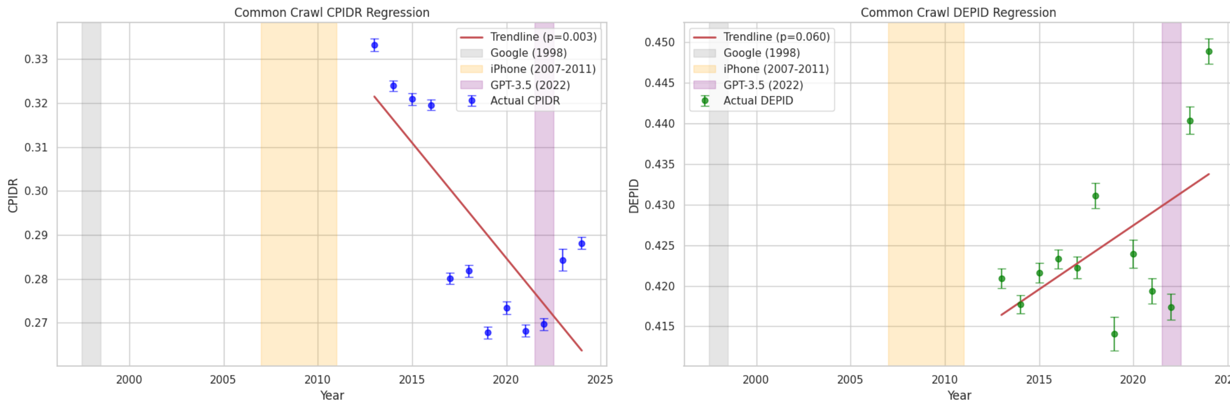
(c) Reddit DEPID distributional analysis (SNR = 0.117, weak signal). Yearly distributions are nearly indistinguishable.

Figure 5.1: Reddit corpus: longitudinal PID trend and distributional analyses.

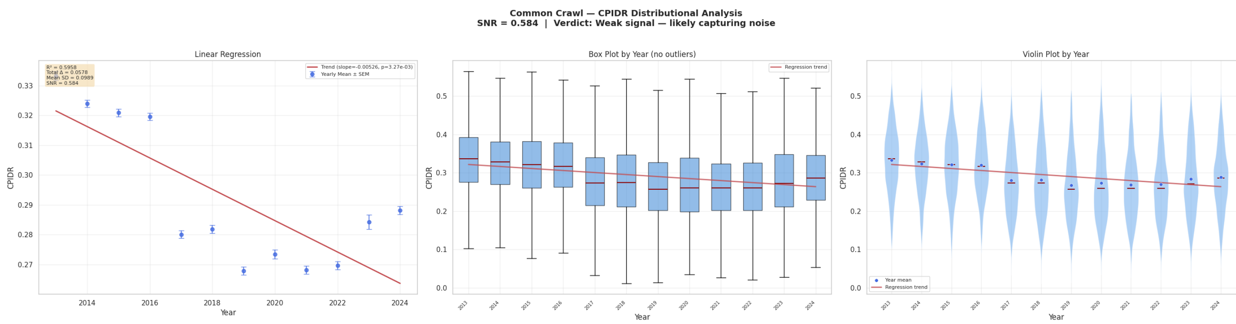
5.3.3 Reddit comments

The Reddit corpus exhibits a negative CPIDR trend ($r = -0.5261$, $p = 0.030$, slope $= -0.000952$/yr) and a directionally concordant DEPID trend ($r = -0.4723$, $p = 0.055$, slope $= -0.000364$/yr) that narrowly fails significance. However, the distributional analysis reveals that both metrics are overwhelmed by within-year noise: CPIDR achieves SNR = 0.106 and DEPID achieves SNR = 0.117, among the lowest in the study. The total predicted CPIDR change over 16 years (0.0152) is less than one-ninth of the mean within-year standard deviation (0.1441). The box and violin plots confirm that yearly distributions overlap almost entirely, with the regression trendline sitting well within the interquartile range of any single year.

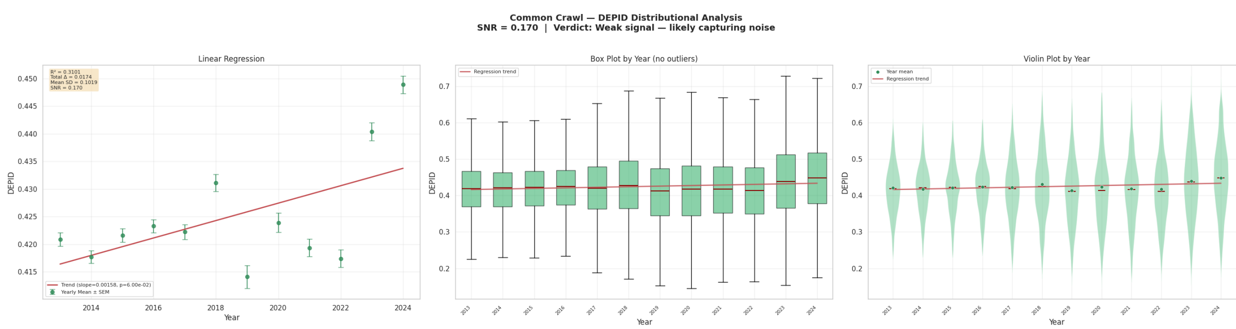
Despite the weak practical signal, the convergence of both metrics toward negative trajectories is exploratorily noteworthy. Several platform-level confounds plausibly contribute to the observed decline in mean scores: explosive user growth coinciding with smartphone adoption around 2012, the "upvote" feedback architecture favoring terse content over discursive argumentation, demographic broadening diluting early-adopter technical communities, and the proliferation of image-macro culture and link aggregation with minimal commentary. These structural forces mechanically suppress both propositional density and syntactic complexity. However, given the extremely low SNR, it is not possible to distinguish a genuine population-level linguistic shift from compositional noise driven by these platform-specific factors.



(a) PID longitudinal trend of Common Crawl dataset.



(b) Common Crawl CPIDR distributional analysis (SNR = 0.584, weak signal). Despite the steepest slope in the study, within-year variance dominates.



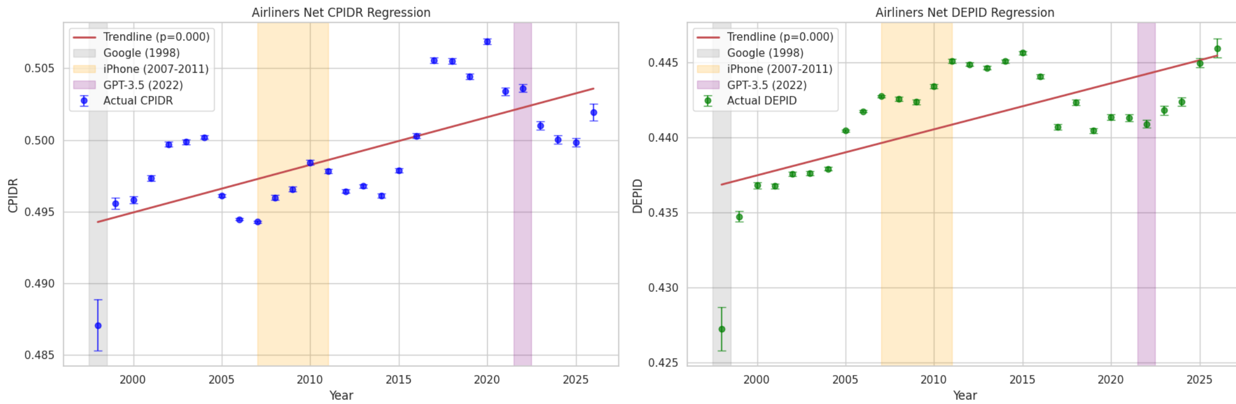
(c) Common Crawl DEPID distributional analysis (SNR = 0.170, weak signal).

Figure 5.2: Common Crawl corpus: longitudinal PID trend and distributional analyses.

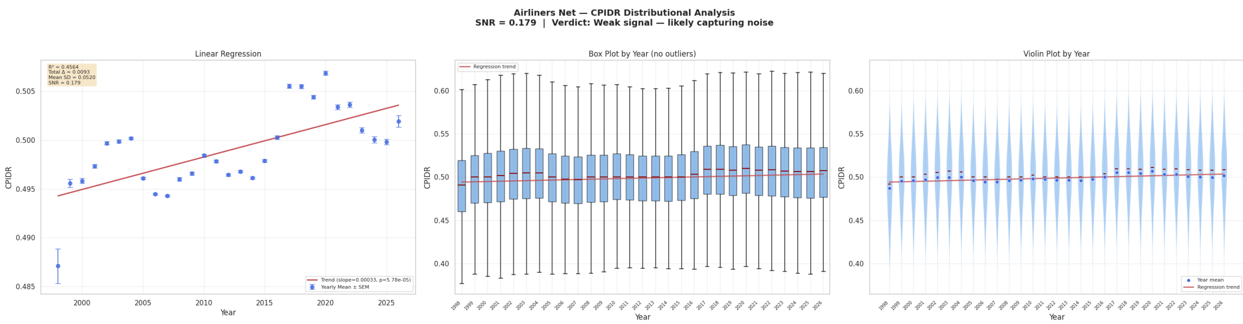
5.3.4 Common Crawl

The Common Crawl corpus yields the steepest absolute CPIDR slope (-0.005255/yr, $r = -0.7718$, $p = 0.003$), while the DEPID trajectory diverges with a positive slope ($r = 0.5568$, $p = 0.060$) that narrowly fails significance. The distributional analysis tempers both findings: CPIDR achieves SNR $= 0.584$ and DEPID achieves SNR $= 0.170$, both classified as weak signals. The mean within-year CPIDR standard deviation (0.0989) dwarfs the total predicted change (0.0578), and DEPID's within-year spread (0.1019) is six times its predicted change (0.0174).

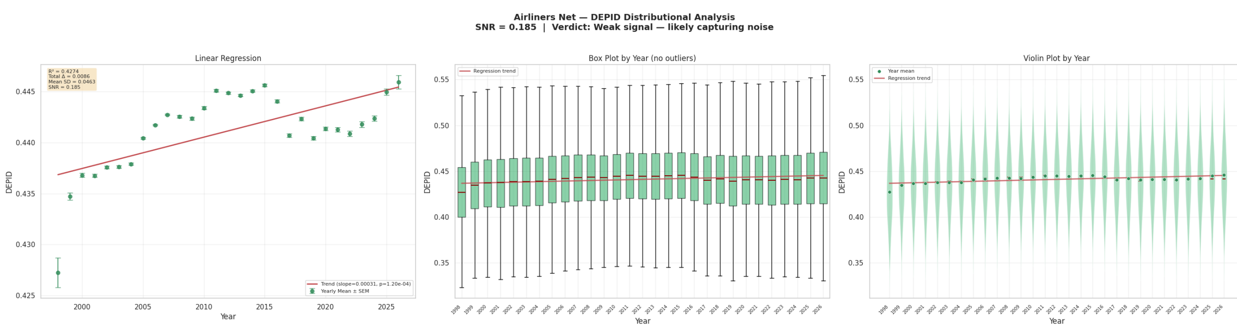
The CPIDR–DEPID dissociation is nonetheless exploratorily interesting. The steep CPIDR slope may reflect structural transformations in the composition of the indexable web: the proliferation of content-farm articles, SEO-optimized landing pages, and mobile-first short-form content systematically reduce the propositional density of crawled documents. The countertrend in DEPID may reflect a compositional shift in which trivial low-density pages multiply while the surviving substantive web corpus concentrates denser syntactic structures. However, the extremely low SNR values indicate that these patterns are not distinguishable from noise at the individual-document level, and this corpus is particularly confounded by heterogeneous source composition changes over time.



(a) PID longitudinal trend of Airliners.net all fora comments.



(b) Airliners.net CPIDR distributional analysis (SNR = 0.179, weak signal). Yearly distributions are nearly indistinguishable despite significant regression.



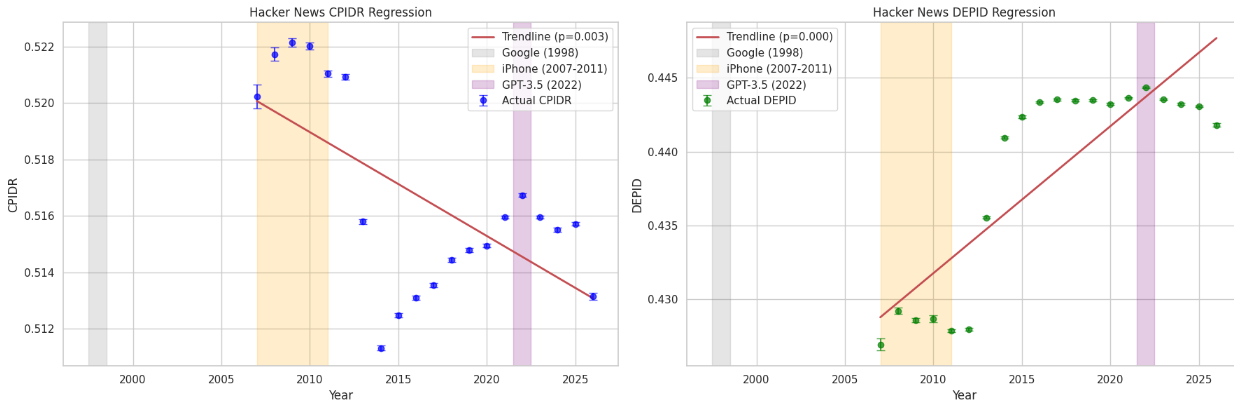
(c) Airliners.net DEPID distributional analysis (SNR = 0.185, weak signal).

Figure 5.3: Airliners.net corpus: longitudinal PID trend and distributional analyses.

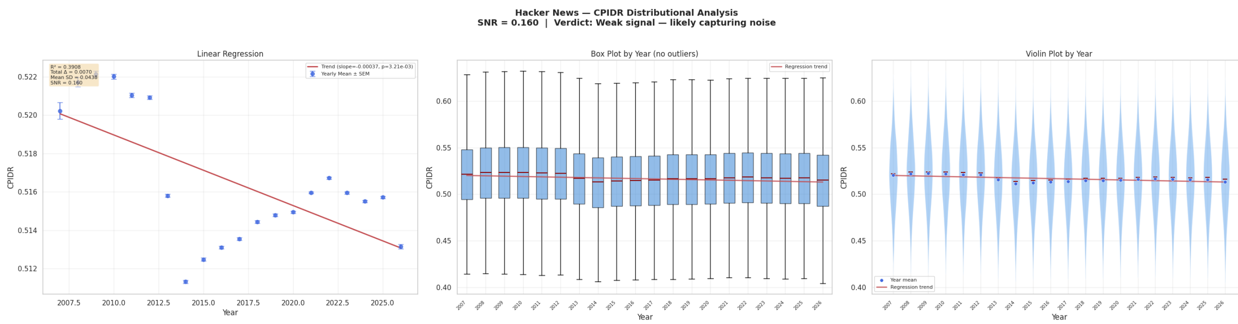
5.3.5 Airliners.net all fora comments

Airliners.net is one of two corpora where both PID metrics achieve significance in the same positive direction: CPIDR ($r = 0.6756$, $p = 5.78 \times 10^{-5}$, slope $= 0.000332$/yr) and DEPID ($r = 0.6537$, $p = 1.20 \times 10^{-4}$, slope $= 0.000306$/yr). However, distributional analysis reveals both as weak signals: CPIDR SNR $= 0.179$ and DEPID SNR $= 0.185$. The total predicted CPIDR change over 28 years (0.0093) is less than one-fifth of the mean within-year standard deviation (0.0520). Box and violin plots show that the yearly distributions are nearly indistinguishable from one another.

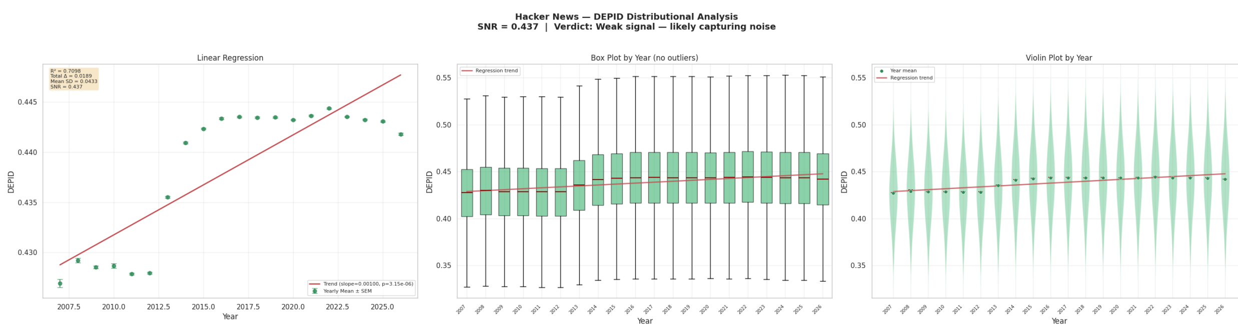
The co-directional significance of both metrics is exploratorily noteworthy, even if the effect sizes are practically negligible. The aviation enthusiast community represents a knowledge-compounding forum with exceptionally low user churn, self-selected domain expertise, and discursive norms established before the smartphone era. The near-parallel slopes may reflect cumulative community maturation, increasing domain complexity (advances in airframe materials, ETOPS routing, regulatory frameworks), and a gradual shift toward more professionally active contributors. However, given SNR < 0.2, these remain speculative explanations for a signal that is indistinguishable from natural within-year variation at the document level.



(a) PID longitudinal trend of Hacker News all fora comments.



(b) Hacker News CPIDR distributional analysis (SNR = 0.160, weak signal).



(c) Hacker News DEPID distributional analysis (SNR = 0.437, weak signal). Despite the highest Pearson r in the study, the trend is less than half the within-year spread.

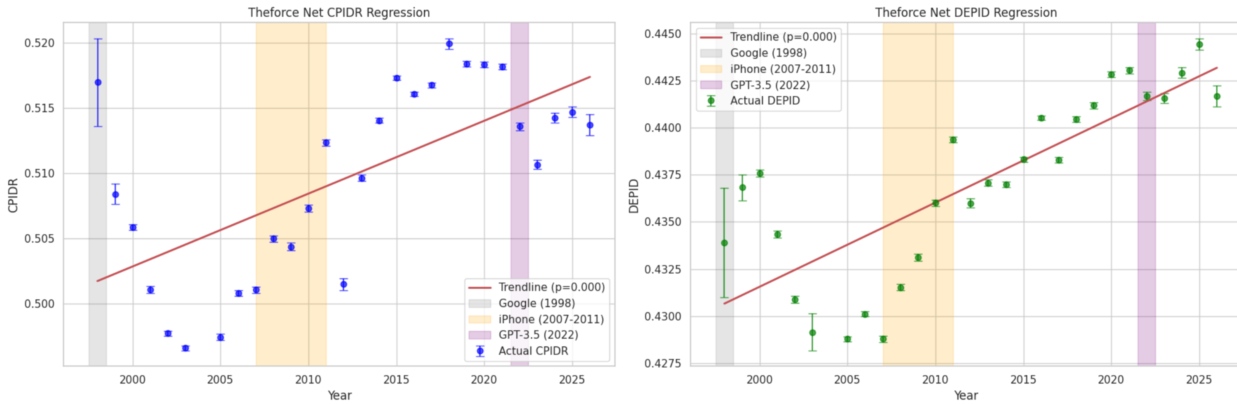
Figure 5.4: Hacker News corpus: longitudinal PID trend and distributional analyses.

5.3.6 Hacker News all fora comments

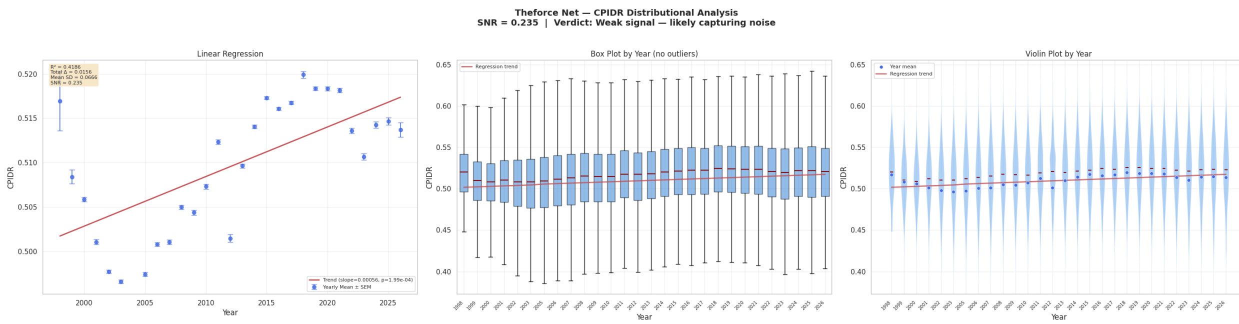
Hacker News produces the starkest CPIDR–DEPID divergence: a significant negative CPIDR trend ($r = -0.6251$, $p = 0.003$, slope $= -0.000368$/yr) coexists with the strongest positive DEPID signal in the study ($r = 0.8425$, $p = 3.15 \times 10^{-6}$, slope $= 0.000997$/yr). The distributional analysis classifies both as weak: CPIDR SNR $= 0.160$ and DEPID SNR

= 0.437. Even the DEPID trend, which achieves the highest Pearson coefficient of any regression in this study, produces a total predicted change over 19 years (0.0189) that is less than half the mean within-year standard deviation (0.0433).

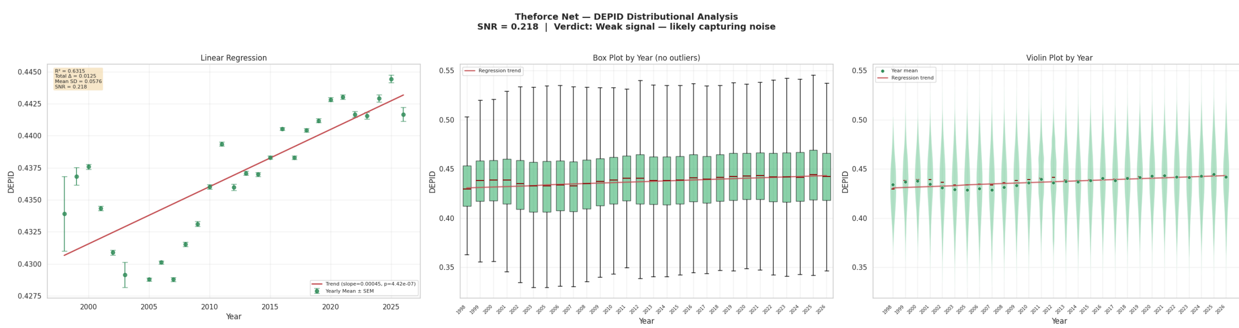
The opposing directionality of the two metrics is nonetheless exploratorily informative. The declining CPIDR may reflect compositional dilution as the platform expanded from a small technical community to a broader audience, increasing the proportion of brief affirmative comments and one-line exchanges relative to the founding epoch of extended technical treatises. The ascending DEPID may indicate that the substantive contributions are becoming syntactically more elaborate, with longer dependency chains driven by increasingly technical topics. This divergence illustrates that CPIDR and DEPID capture genuinely orthogonal dimensions of linguistic complexity, and no single PID metric can serve as a universal proxy. However, the weak SNR for both metrics means these directional observations should be treated as hypotheses for future investigation rather than confirmed trends.



(a) PID longitudinal trend of TheForce.net all fora comments.



(b) TheForce.net CPIDR distributional analysis (SNR = 0.235, weak signal).



(c) TheForce.net DEPID distributional analysis (SNR = 0.218, weak signal). Heavily overlapping violins across 28 years.

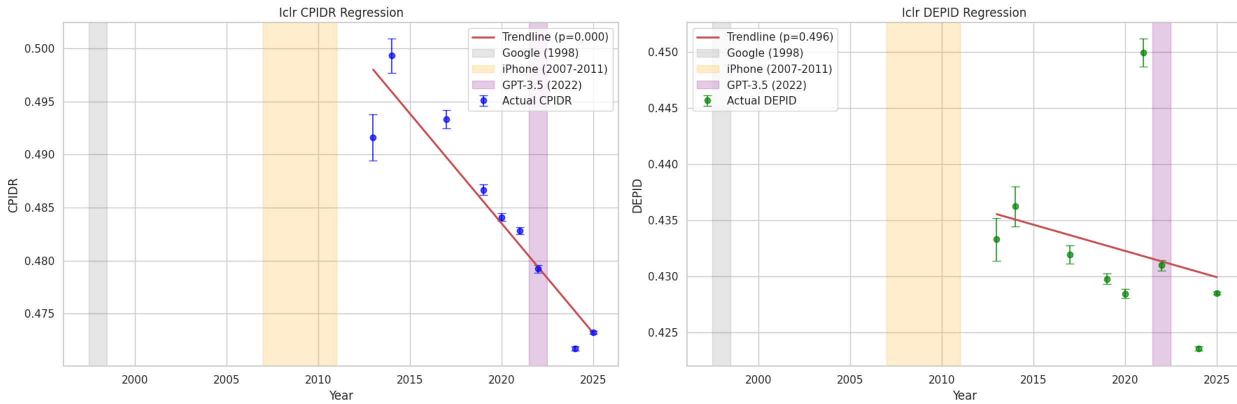
Figure 5.5: TheForce.net corpus: longitudinal PID trend and distributional analyses.

5.3.7 TheForce.net all fora comments

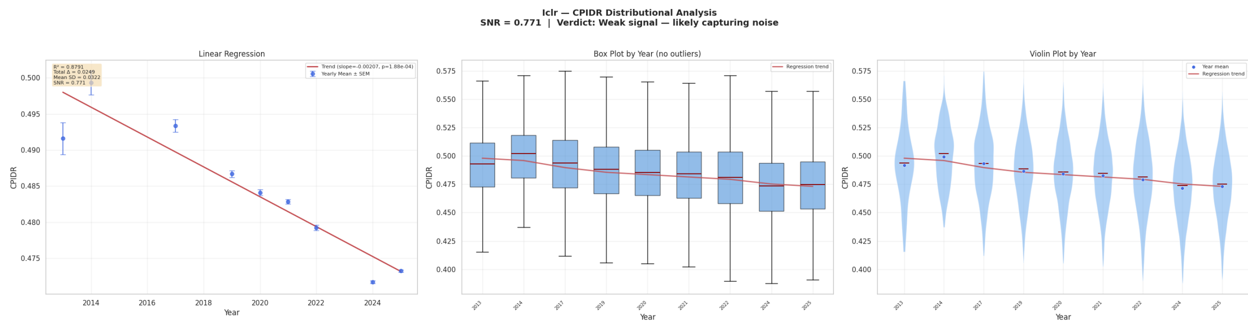
TheForce.net achieves among the strongest p-values in the study for both metrics: CPIDR ($r = 0.6470$, $p = 1.98 \times 10^{-4}$, slope $= 0.000559$/yr) and DEPID ($r = 0.7946$, $p = 4.42 \times 10^{-7}$, slope $= 0.000447$/yr). Yet the distributional analysis classifies both as weak signals: CPIDR SNR $= 0.235$ and DEPID SNR $= 0.218$. The total predicted DEPID change over 28 years

(0.0125) is roughly one-fifth of the mean within-year standard deviation (0.0576), and year-by-year violin plots show heavily overlapping distributions.

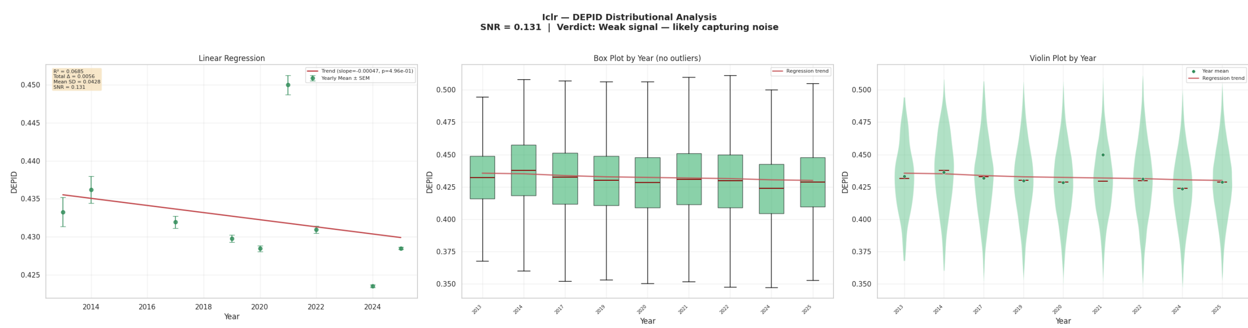
This corpus nonetheless represents a valuable natural experiment: a closed, demographically stable fan community organized around a discrete fictional universe, with minimal platform disruption and no algorithmic feed curation. The co-directional positive trends in both metrics over more than two decades are consistent with a hypothesis of cumulative community maturation: as the Star Wars expanded universe grew in complexity across The Clone Wars, Rebels, and the sequel trilogy, the syntactic architecture required for precise intra-universe argumentation may have become more demanding. Whether this constitutes genuine linguistic enrichment or an artifact of changing community composition cannot be resolved by the current SNR analysis alone and requires further investigation.



(a) PID longitudinal trend of ICLR peer-review comments.



(b) ICLR CPIDR distributional analysis (SNR = 0.771, weak signal). Despite $R^2 = 0.879$, the total predicted change is smaller than within-year SD.



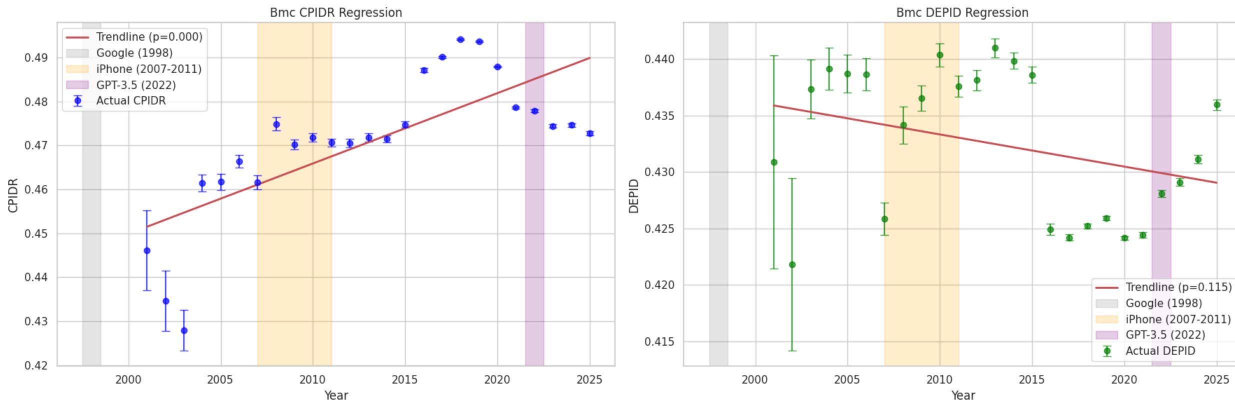
(c) ICLR DEPID distributional analysis (SNR = 0.131, weak signal).

Figure 5.6: ICLR corpus: longitudinal PID trend and distributional analyses.

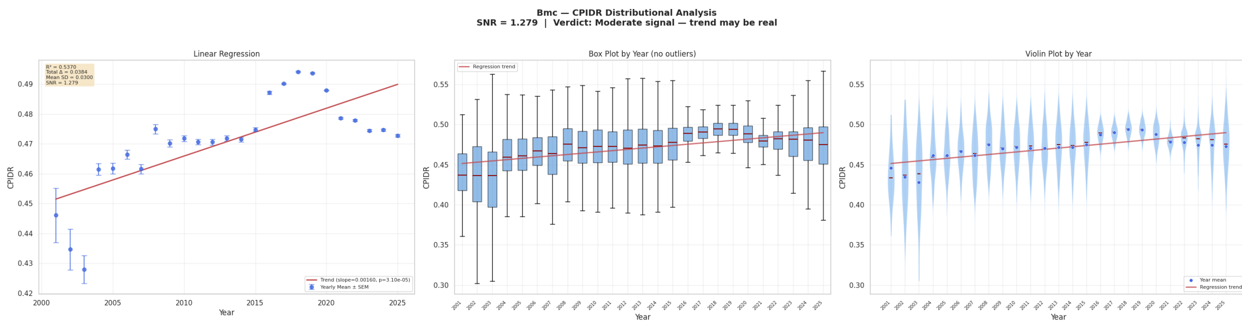
5.3.8 ICLR peer-review comments

ICLR yields the strongest single regression in the study: CPIDR ($r = -0.9376$, $p = 1.88 \times 10^{-4}$, slope $= -0.002071$/yr), a near-perfect linear decline. DEPID is non-significant ($r = -0.2618$, $p = 0.496$). Despite the exceptional regression fit, the distributional analysis classifies CPIDR as a weak signal (SNR = 0.771) and DEPID as weak (SNR = 0.131). The total predicted CPIDR change over 12 years (0.0249) is smaller than the mean within-year standard deviation (0.0322).

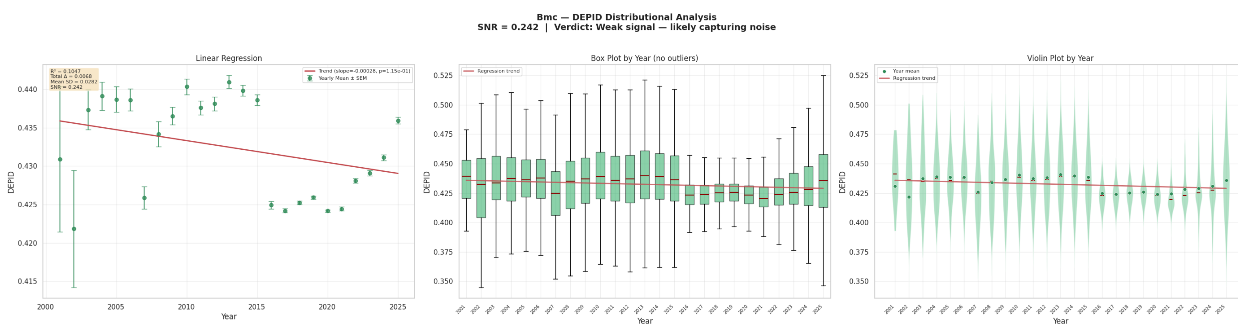
The near-perfect $R^2 = 0.879$ coexisting with a weak SNR is itself an instructive methodological observation: it demonstrates that regression on aggregated means can achieve extremely high goodness-of-fit even when the underlying document-level distributions overlap almost entirely. The ICLR case is the clearest illustration of why p-values and R^2 alone are insufficient for assessing practical significance. Exploratorily, the declining CPIDR mean may reflect the exponential expansion of the reviewer pool (incorporating more early-career and non-native English writers), the shift toward template-driven review formats, and potentially the early influence of LLM-assisted writing post-2022. The absence of a DEPID trend suggests that whatever is reducing propositional counts is not simplifying grammatical structure. These remain hypotheses warranting dedicated study rather than confirmed conclusions.



(a) PID longitudinal trend of BMC peer-review comments.



(b) BMC CPIDR distributional analysis (SNR = 1.279, moderate signal). The only corpus-metric pair where the trend magnitude is comparable to within-year variability.



(c) BMC DEPID distributional analysis (SNR = 0.242, weak signal).

Figure 5.7: BMC corpus: longitudinal PID trend and distributional analyses.

5.3.9 BMC peer-review comments

BMC is the only corpus achieving a moderate SNR for any metric: CPIDR yields SNR = 1.279, the highest value in the study ($r = 0.7328$, $p = 3.10 \times 10^{-5}$, slope = 0.001601/yr). The total predicted CPIDR change over 24 years (0.0384) exceeds the mean within-year standard deviation (0.0300), making this the sole corpus-metric pair where the longitudinal trend magnitude is comparable to within-year variability. DEPID remains a weak signal ($r = -0.3236$, $p = 0.114$, SNR = 0.242).

The moderate CPIDR signal in biomedical peer review is the most promising candidate for a genuine trend in this study. The increasing complexity and multidisciplinarity of life-science research over the past two decades has imposed growing propositional demands on reviewers, who must integrate critique across genomics, biostatistics, clinical trial design, and bioinformatics. The normalization of open peer review within the BMC portfolio may have progressively elevated propositional density standards, and structured reporting compliance (CONSORT, PRISMA, STROBE) further loads propositional content. The absence of a DEPID increase may reflect a stylistic shift toward syntactically streamlined, propositionally dense construction: a hallmark of modern biomedical register. While this is the strongest practical signal observed, it remains moderate (SNR = 1.279), and future work should investigate whether this trend replicates across other biomedical journal families.

5.4 Conclusion

This exploratory longitudinal analysis demonstrates that while statistically significant regressions in PID scores can be detected across massive digital text archives, the distributional analysis reveals that nearly all observed trends are dominated by within-year variance. Of 14 corpus-metric regressions, none achieves a strong signal-to-noise ratio (SNR > 2), and only BMC CPIDR reaches a moderate level (SNR $= 1.279$). The remaining 13 regressions are classified as weak signals (SNR ≤ 1), meaning the total predicted change over multi-decade observation windows is smaller than the typical standard deviation of PID scores among documents within any single year.

5.5 Discussion and Future Work

The central methodological lesson of this exploratory analysis is that statistical significance on aggregated yearly means does not imply practical significance at the document level. With sample sizes exceeding 12.6 million documents, extremely small shifts in population means become statistically detectable, but these shifts are dwarfed by the natural variability of individual documents. This finding does not invalidate the observed directional patterns, the divergent trajectories across corpora remain exploratorily informative, but it does preclude definitive causal attribution to cognitive decline, platform culture, or any other single factor.

The exploratory observations nonetheless generate several testable hypotheses for future work. The co-directional positive trends in domain-specific communities (Airliners.net, TheForce.net) and the CPIDR–DEPID dissociations in Hacker News and Common Crawl suggest that platform culture and domain complexity may influence longitudinal PID trajectories, but establishing this will require methods that go beyond aggregate regression, such as within-user longitudinal tracking, matched cohort analyses, or causal inference designs that control for compositional confounds. The moderate BMC CPIDR signal (SNR $= 1.279$) is the most promising candidate for replication across additional biomedical journal families. Furthermore, this vast longitudinal tracking acts as a preliminary, foundational framework for a vastly expanded suite of cognitive research. Empowered by the psycholinguistic precedents established by the original Nun Study [1], future iterations of this methodology will be capable of analyzing substantially more longitudinal text formats and datasets. By combining these traditional linguistic density markers with highly scalable, state-of-the-art diagnostic architectures, we establish a foundation for continuously monitoring digital cognitive health trajectories across emerging communication domains in future studies.

Chapter 6

Conclusion

6.1 Summary

This thesis was inspired by the pioneering work of Dr. David Snowden and the Nun Study, which established a profound statistical association between early-life linguistic ability, measured by Propositional Idea Density (PID), and the later-life incidence and severity of Alzheimer's Disease. Snowden's discovery that idea density scores from autobiographies written at a mean age of 22 exhibited strong inverse correlations with neurofibrillary tangle counts decades later motivated the central premise of this work: that linguistic ability, quantified through both classical rule-based metrics and modern deep learning architectures, can serve as a meaningful signal of cognitive health at both the individual and population level. Building on this foundation, we formulated two research questions and conducted extensive experiments to investigate (1) whether Transformer-based Artificial Neural Networks can attend to linguistic patterns associated with cognitive decline, and (2) whether population-scale linguistic ability trends can be demonstrated through longitudinal PID analysis across diverse text corpora.

6.2 Answering the Research Questions

6.2.1 Research Question 1: Can Transformer-based ANNs attribute relevant linguistic patterns to predict an individual's cognitive decline?

Our experiments on the DementiaBank Pitt Corpus provide a definitive affirmative answer, with important qualifications. We first established that the three rule-based PID metrics: CPIDR, DEPID, and DEPID-R, are insufficiently sensitive to discriminate between Dementia and Control transcripts in conversational speech as depicted in the Pitt corpus, with none achieving statistical significance at $\alpha = 0.05$. This finding, while seemingly at odds with Snowden's original results, is consistent with the critical distinction between formal autobiographical writing and semi-structured interview transcripts: the latter introduce substantial noise from interviewer prompts, task-specific vocabulary, and conversational artifacts that dilute propositional density signals.

Non-LLM machine learning models (TF-IDF XGBoost, MLP, 1D CNN, BiLSTM with Attention) achieved deceptively high performance up to perfect F1 = 1.000. However, systematic interpretability analysis via native feature importances, Permutation Importance, and Integrated Gradients revealed pervasive spurious correlation memorization instead of generalization. These models exploited task-specific vocabulary from the Cookie Theft elicitation protocol and idiosyncratic conversational tangents rather than learning generalizable markers of cognitive decline. Notably, the MLP and BiLSTM models achieved perfect scores precisely because their TF-IDF feature spaces maximally expose the spurious vocabulary correlations that enable memorization.

Pre-trained Transformer-based encoders, operating on sub-word tokenizations decoupled from the task-specific vocabulary, successfully mitigated this effect. A consistent empirical pattern emerged across all four encoder architectures (DistilBERT, RoBERTa, ModernBERT, and Llama-Embed-Nemotron-8B): fine-tuned models uniformly outperformed their frozen encoder counterparts, demonstrating that supervised domain adaptation is essential for maximizing predictive performance even in data-scarce clinical settings. The **RoBERTa FT + GRU sliding window architecture** emerged as the strongest configuration, achieving perfect classification (macro F1 = 1.000, accuracy = 1.000), with RoBERTa Full Fine-Tuning closely following (macro F1 = 0.963). RoBERTa's 125M parameter count occupies an optimal trade-off between representational capacity and adaptability: large enough to capture nuanced semantic features, yet small enough for stable full fine-tuning on 234 training samples without catastrophic overfitting. Llama-Embed-Nemotron-8B, despite its 8B parameters, achieved a matching ceiling under LoRA (F1 = 0.963), demonstrating that parameter-efficient fine-tuning can be competitive but does not surpass the smaller, fully fine-tuned encoder on this data regime.

Critically, SHAP interpretability analysis comparing the frozen and fine-tuned RoBERTa models confirmed that fine-tuning amplifies the encoder's sensitivity to theoretically grounded linguistic markers while suppressing attention to dataset-specific vocabulary artifacts. The fine-tuned model's attributions concentrate on propositional density and specific descriptive vocabulary in Control transcripts (action verbs and concrete nouns), and on conversational fillers, vague pronominal references, and fragmented syntactic structures in Dementia transcripts, precisely the markers that clinical linguistics associates with reduced propositional idea density and cognitive decline. The RoBERTa + GRU sliding window architecture further introduced a hierarchical interpretability mechanism, with per-window attention weights revealing which transcript segments the model considers most diagnostic. Thus, Transformer-based ANNs do not merely predict cognitive decline; they attend to the same underlying linguistic dimensions that PID was designed to measure, but with far greater sensitivity and robustness to the noise inherent in transcribed speech.

An additional finding emerged from our investigation into attention-weighted propositional idea density. While rule-based PID metrics fail to discriminate at the whole-document level, DEPID-R, which penalizes propositional repetition, a cardinal symptom of Alzheimer's disease, approached marginal significance ($r_{pb} = -0.099$, $p = 0.090$), suggesting that repetition-aware density measures may capture a clinically meaningful signal that standard PID averaging dilutes. When the fine-tuned RoBERTa + GRU model's learned window-attention weights were used to re-weight per-window PID scores, the association strengthened: CPIDR achieved Mann-Whitney $p = 0.034$ on the multi-window subset under softmax-sharpened attention weighting. Although most parametric tests remained marginally non-significant (Welch's t-test $p = 0.060$), the experiment demonstrated that the model's attention can weights carry information about *where* in a transcript cognitive deficits are most acute, and that this localization signal can be leveraged to bridge neural classification and traditional linguistic metrics. This motivates a hybrid paradigm in which hierarchical attention architectures serve as weighting functions over clinically interpretable rule-based metrics computed at the sub-document level: an approach that the present study can only motivate but that larger, more balanced clinical datasets may validate.

6.2.2 Research Question 2: Can we demonstrate a linguistic ability trend for a specific population of writers based on the PID score of their comments?

The longitudinal trend analysis across seven massive text corpora, spanning up to 29 years and encompassing over 12.6 million documents, produced nuanced and cautionary results. OLS regression with Pearson correlation analysis revealed statistically significant trends in most corpora: all seven CPIDR regressions achieved $|r| \geq 0.5$ with $p \leq 0.05$, and three DEPID regressions (Airliners.net, Hacker News, TheForce.net) achieved significance with positive trends.

However, a subsequent distributional analysis fundamentally reframed these findings. By computing a Signal-to-Noise Ratio (SNR), defined as the total predicted change over the observation window divided by the mean within-year standard deviation. We found that **none** of the 14 corpus-metric regressions achieved a strong signal (SNR > 2). Only BMC CPIDR reached a moderate level (SNR $= 1.279$), and the remaining 13 were classified as weak signals (SNR ≤ 1). This means that for virtually every corpus, the total predicted change in PID over the entire multi-decade window is *smaller* than the typical variation among documents within a single year. For example, Hacker News DEPID achieves the strongest Pearson correlation in the study ($r = 0.8425$, $p = 3.15 \times 10^{-6}$), yet its total predicted change over 19 years (0.0189) is less than half the mean within-year standard deviation (0.0433).

The answer to RQ2 is therefore qualified: statistically significant longitudinal trends in PID scores can be detected when aggregating over sufficiently large corpora, but the practical significance of these trends is negligible relative to the natural variability of individual documents. The regression on yearly means is reliable in detecting that the population mean shifted, but the magnitude of the shift is so small that it cannot be confidently attributed to any single cause and may not be meaningful.

6.3 Broader Implications

The convergence of findings from both research questions yields several implications for the fields of clinical linguistics, computational cognitive science, and AI-assisted health assessment.

First, rule-based PID metrics, while foundational to the Nun Study's discoveries, are insufficient for individual-level cognitive decline prediction on conversational speech transcripts. Their value at the population level is also more limited than initially hypothesized: our distributional analysis demonstrates that even with millions of documents, the longitudinal trends in PID scores are overwhelmed by within-year variance (SNR ≤ 1 for 13 of 14 regressions). This suggests that PID-based population monitoring requires methods more sophisticated than aggregate linear regression, such as within-user longitudinal tracking or matched cohort designs to yield practically meaningful conclusions. Individual-level diagnosis should leverage the richer representational capacity of pre-trained language models.

Second, the consistent superiority of fine-tuned encoders over frozen variants underscores that even in data-scarce clinical settings, supervised domain adaptation through fine-tuning successfully specializes pre-trained representations to the clinical task. RoBERTa (125M parameters) achieved the largest fine-tuning gain (ΔF1 $= +0.072$) and the strongest non-GRU performance (F1 $= 0.963$), while the 64$\times$ larger Llama-Embed-Nemotron-8B matched this ceiling under LoRA but with a smaller improvement margin (ΔF1 $= +0.019$). This suggests that for clinical NLP with limited labeled data, moderately-sized encoders occupy an optimal point in the bias-variance trade-off, and that the marginal returns of scaling model parameters diminish rapidly when training data is scarce.

Third, the exploratory longitudinal analysis cautions against interpreting statistically significant regressions on aggregated means as evidence of meaningful real-world trends. With sample sizes exceeding 12.6 million documents, extremely small population-mean shifts become detectable, but these shifts are dwarfed by natural document-level variability. The coexistence of rising and falling PID trajectories across communities sharing the same temporal window, technological environment, and broad demographic context suggests that platform culture, domain complexity, and communicative norms are plausible confounds, but the weak SNR values preclude definitive causal attribution. Apparent declines in linguistic complexity on mass platforms should not be reflexively interpreted as evidence of cognitive deterioration; equally, apparent increases in niche communities should not be interpreted as evidence of cognitive enrichment.

6.4 Future Work

Several directions emerge naturally from this work. The most immediate priority is the application of the fine-tuned encoder methodology to the Nun Study autobiographies, should the data become available. Training on both written autobiographies and transcribed speech corpora would expose models to multiple registers of cognitively impaired language, potentially enabling the cross-modal generalization that the Pitt Corpus alone cannot support. Despite achieving perfect classification on the Pitt Corpus test set (macro F1 = 1.000 for the RoBERTa FT + GRU architecture), the models' reliance on speech-specific artifacts indicates a fundamental data diversity bottleneck; substantially larger and more diverse clinical corpora, spanning structured clinical interviews, free-form narrative tasks, and written correspondence across multiple elicitation protocols, will be required for these models to generalize across elicitation modalities and detect cognitive decline regardless of the specific linguistic task.

A second priority, motivated by our attention-weighted PID analysis, is the systematic investigation of the hybrid neural–traditional metric paradigm on multi-site clinical corpora. The present study demonstrated that attention-weighted aggregation of per-window PID scores moves the association with dementia in the correct direction, but the severely limited Control sample in the multi-window subset ($n = 12$) precluded definitive validation. Larger datasets with balanced representation of long Control transcripts would provide the statistical power needed to determine whether attention-weighted DEPID-R, which directly penalizes the perseverative propositional repetition characteristic of Alzheimer's disease, can aid diagnosis with interpretable prediction. If validated, this approach would combine the clinical interpretability and theoretical grounding of rule-based PID with the localization precision of deep learning attention mechanisms, offering clinicians both a prediction and a principled indication of which transcript regions warrant closer examination.

For the longitudinal analysis, the most critical methodological advance needed is a transition from aggregate regression on yearly means to within-user longitudinal tracking. The weak SNR values observed across all corpora demonstrate that aggregate trends, while statistically detectable, are practically negligible relative to document-level variance. Future work should track individual users' PID trajectories over time, controlling for user tenure, topic category, and platform-specific engagement metrics, to determine whether within-person trends are stronger than the between-person noise that dominates the current analysis. Matched cohort analyses comparing users who joined before and after specific technological inflection points (smartphone adoption, LLM availability) would provide stronger potential causal evidence than cross-sectional regression.

The moderate BMC CPIDR signal (SNR = 1.279) warrants dedicated replication across additional biomedical journal families to determine whether this is a genuine domain-specific trend or a statistical artifact. The CPIDR decline observed in ICLR peer reviews, while classified as a weak signal (SNR = 0.771), remains exploratorily interesting given the potential influence of LLM-assisted writing on scientific discourse density post-2022. Recent neuroscientific evidence underscores this concern: Kosmyna et al. [60] demonstrated that participants who used ChatGPT for essay writing over multiple sessions exhibited progressively weaker EEG connectivity, reduced cognitive engagement, and diminished linguistic ownership compared to unaided writers, a phenomenon the authors term *cognitive debt*. Whether habitual LLM reliance causally reduces human idea density remains an open question with clinical implications, given PID's standing with respect to written autobiographical text as a predictor of neurodegenerative disease decades before clinical onset.

Ultimately, this thesis establishes that linguistic ability, as quantified by propositional idea density, remains a powerful lens through which to examine cognitive health for individuals through deep learning architectures that capture the semantic and pragmatic dimensions of language that rule-based metrics cannot. The exploratory longitudinal analysis, while demonstrating that population-level PID trends are dominated by within-year noise under current methodology, provides a foundational framework and identifies specific hypotheses for future investigation using more granular analytical approaches. By combining these classical linguistic markers with scalable, state-of-the-art diagnostic architectures, we establish a potential foundation for continuously monitoring digital cognitive health trajectories across emerging communication domains in future studies.

APPENDIX A: DATA USE ACKNOWLEDGMENT AND IRB APPROVAL

DementiaBank Pitt Corpus

We are grateful for the opportunity to use data from the DementiaBank Pitt Corpus, [18] which was collected at the University of Pittsburgh School of Medicine from participants recruited between 1983 and 1988 as part of a longitudinal study of the natural history of Alzheimer's disease. We acknowledge and thank the researchers at the University of Pittsburgh Alzheimer's Disease Research Center who designed the study protocol, conducted the clinical interviews, and transcribed the speech data that comprise this corpus. We also wish to express our sincere appreciation to the participants and their families who generously volunteered their time and contributed to these interviews. Their willingness to participate in clinical research is invaluable to advancing our understanding of Alzheimer's disease and cognitive decline.

IRB Approval

Our use of the DementiaBank Pitt Corpus data in this thesis was reviewed and approved by the Southern Methodist University Institutional Review Board (IRB) under Protocol ID 25-184. The review type was Exemption, and the approval status is Exemption Verified under Category (4): Secondary Research Uses of Data or Specimens.

APPENDIX B: USE OF AI TOOLS

The following disclosure details the use of artificial intelligence tools during the preparation of this dissertation.

Tools Used

1. **Claude Opus 4.6** (Anthropic): Used for (a) generating LaTeX tables from raw experimental results and (b) grammar and spelling review of drafted text.

2. **Grammarly** (Grammarly, Inc.): Used for grammar and spelling review of drafted text.

Scope of Use

All AI-assisted outputs were reviewed, verified, and edited by the author prior to inclusion. Specifically:

- **Table generation**: Experimental results (e.g., regression statistics, model performance metrics) were computed independently by the author's code. Claude Opus 4.6 was used solely to format these pre-computed numerical results into LaTeX table markup. All values were verified against the original computational outputs.

- **Grammar and spelling**: Both Claude Opus 4.6 and Grammarly were used as proofreading aids. All suggested corrections were reviewed and accepted or rejected at the author's discretion. Neither tool was used to generate substantive intellectual content, analytical arguments, or scientific interpretations.

Attestation

All research design, data collection, computational experiments, statistical analyses, scientific interpretations, and scholarly arguments presented in this dissertation are the original intellectual work of the author. The AI tools listed above served exclusively as formatting and proofreading utilities.

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